Public Displays of Alignment*

Zi Yang Kang† Joris Mueller‡ Jaya Wen§ Cheryl Wu¶

August 2026

Abstract

Why do firms publicly echo government language? Two common explanations are *noise*, under which alignment is payoff-irrelevant, and *government consumption*, under which the regime rewards aligned firms because it directly values alignment. We propose a third explanation: the regime can use alignment to *screen* firms for costly political support. We develop a contracting framework encompassing all three explanations and derive two results. First, absent screening, alignment cannot predict differences in firms' mean payoff responses or costly actions when political stress materializes. Second, under optimal screening, lower-cost firms align, provide greater support, and receive more favorable treatment. We test these predictions using a transparent measure constructed from listed Chinese firms' use of regime-specific phrases in annual reports. More-aligned firms experience larger valuation losses after an adverse political shock and provide costly support by retaining more workers during local labor unrest. Within our framework, these responses rule in a screening role and are consistent with optimal screening.

JEL classification: D02, D82, G38, P16, P26

* We thank Zehang Chen, Hui Yann Foo, Hana Kwon, Jun Yang Liew, Sandy Liu, Hans Neddyanto Tandjung, En Qi Teo, Jinghong Wang, Johnny Lee Zhuang Yu, and especially Junyi Liu and Chuyue Tian, for excellent research assistance. We thank Jijian Fan for sharing data on political connections. We are grateful to Richard Bluhm, Gabriel Carroll, Simon Evenett, Ray Fisman, Gordon Hanson, Chang-Tai Hsieh, Ruixue Jia, Wolfgang Keller, Elisabeth Kempf, Wendy Leutert, Vincent Pons, Nancy Qian, Meg Rithmire, Jesse Shapiro, Xianwen Shi, Andy Skrzypacz, Tuan-Hwee Sng, Michael Song, Colin Stewart, David Strömberg, Marco Tabellini, Shosh Vasserman, Shaoda Wang, and David Yang, as well as numerous seminar and conference participants. The authors used AI tools for editorial assistance in manuscript preparation. All analysis, arguments, conclusions, and mistakes are their own. Mueller gratefully acknowledges funding from the Singapore Ministry of Education AcRF Tier 1 grant FY2023-FRC2-006.

† Department of Economics, University of Toronto; zy.kang@utoronto.ca.

‡ Department of Economics, National University of Singapore; jorismueller@nus.edu.sg.

§ Harvard Business School, Harvard University; jwen@hbs.edu.

¶ Department of Economics, Yale University; cheryl.wu@yale.edu.

1 Introduction

Many firms publicly align themselves with governments through corporate speech. In the United States, for example, some companies began adopting pro-Trump language after the 2024 election.[1] Such behavior is especially pronounced in autocracies: firms in China, Russia, and Saudi Arabia routinely incorporate regime slogans into their public communications.[2] Yet despite its prevalence, corporate political speech directed at the government remains theoretically and empirically underexplored. Why do firms engage in public alignment, and what economic function does it serve? These questions address a central problem in the political economy of autocracy, namely how regimes manage firms when formal commitment is weak, information is limited, and political support is valuable (Shleifer and Vishny, 1998; Acemoglu and Robinson, 2005; Svolik, 2012; Egorov and Sonin, 2024).

Two common explanations are *alignment as noise* and *alignment as government consumption*. Under the first, public alignment conveys no payoff-relevant information to the regime. Under the second, the regime directly values public alignment and rewards aligned firms with favorable treatment. As such, *all* firms are incentivized to align.

We introduce a third explanation: *alignment as screening*. Under this explanation, public alignment separates firms according to their expected costs of providing political support. Thus, it helps the regime identify firms that are more willing or able to provide costly support when needed, such as when political conditions deteriorate. Under optimal screening, only lower-cost firms are incentivized to align.

Anecdotal evidence is consistent with all three explanations but does not distinguish among them. Theoretically, it is not obvious how each mechanism arises, how it can be sustained, or what observable implications it generates. Empirically, little is known about which firms align, whether alignment predicts their behavior and treatment by the regime, and whether any such relationships reflect the

[1] See Washington Post, March 13, 2025 and Semafor, June 16, 2025.
[2] See City Cement, 2023 Annual Report; Rosneft, 2024 Annual Report; and Zhongyuan Bank, 2024 Annual Report.

effects of alignment itself or other correlated firm characteristics.

We answer these questions by developing a unified theoretical framework and evaluating its predictions using empirical evidence from China. Our framework accommodates all three explanations and characterizes distinct implications of each. We then construct a general and transparent measure of public alignment using annual reports from listed Chinese firms and use it to document new facts about which firms align, how the government treats aligned firms, and how they respond to periods of political stress. Within our framework, the evidence rules in a screening role for public alignment, rules out noise, and allows that alignment may have consumption value for the government.

Our framework is motivated by a central information problem in autocratic governance. Regimes often rely on firms for support during periods of political stress but may not know which firms can provide it at relatively low cost. Firms have private information about their productivity, organizational capacity, and outside opportunities, all of which affect the opportunity cost of diverting resources toward political objectives. This informational asymmetry is a familiar theme in political economy (Hayek, 1945; Huang et al., 2017; Egorov and Sonin, 2024) and echoes what Yang (2024) describes as a fundamental challenge of governance in contemporary China.

This information problem creates a potential screening role for public alignment. While the regime may directly value both public alignment and costly support, alignment may additionally help the regime identify firms that are more willing or able to provide costly support when political conditions deteriorate. Using this information to allocate support is valuable because coercion is an imperfect substitute for information: forcing support from firms may destroy surplus, deter investment, and reduce the resources on which the regime itself depends. The regime therefore benefits from identifying, without resorting to coercion, firms for which support is less costly. Because public alignment is observable and durable, the regime can condition support burdens and favorable treatment on firms' alignment choices, inducing them to self-select according to their costs of support. Other forms of state embeddedness, such as political connections, ownership ties, or Party organizations, may also screen firms; our focus is on public alignment as a distinct and measurable margin.

Formally, we study a contracting problem between a regime and a firm. The firm privately observes its cost of providing support. Although alignment itself is costless, the regime can attach contractual consequences to the firm's alignment choice: it offers a menu that associates each choice with a support burden and a transfer. The framework nests the three explanations described above. Alignment can arise as noise when it is payoff-irrelevant within the firm–regime relationship, as government consumption when the regime derives direct value from alignment and support, and as screening when alignment separates firms with different support costs.

We use the framework for two purposes. First, we derive a restriction that distinguishes whether alignment has any screening role. Absent screening, alignment cannot predict differential support responses or differential mean firm-payoff responses. Observing either difference thus rules in a screening role. Second, we characterize the surplus-maximizing use of alignment as a screening device. When the surplus-maximizing menu separates firms, it assigns greater support burdens to firms with lower opportunity costs of support and compensates them through favorable treatment. The regime can therefore use alignment to direct support burdens toward firms for which such support is less costly, rather than relying on uniform coercion. This characterization generates predictions about which firms align and yields comparative statics on the value and prevalence of screening.

To evaluate the framework's empirical implications, we develop a transparent text-based measure of public alignment. We use machine-learning methods to identify phrases distinctive to regime documents and then measure their use in firms' annual reports. We apply this procedure to listed Chinese firms, identifying the 50 phrases that most clearly distinguish Chinese Communist Party (CCP) language from standard Mandarin and aggregating the resulting indicators into a firm-year index using their first principal component. Firms whose annual reports contain more regime-specific language receive higher alignment scores.

We assess the measure's validity and robustness in several ways. We vary the method used to aggregate phrase indicators into an index and the portion of each annual report we search. We also show that the measure is substantially correlated with an LLM-based alternative, while remaining

more transparent and easier to interpret. We next show that more than a fifth of reports contain explicit pledges of loyalty or support, indicating that firms use regime-specific language affirmatively rather than simply referring to it neutrally. Finally, the measure is only moderately correlated with existing measures of political risk (Hassan et al., 2019), suggesting that it captures a distinct dimension of firm speech.

We document new facts about how public alignment varies across firms and over time using this measure. First, we find substantial firm-level heterogeneity. Year, industry, city, and industry-by-city fixed effects together explain about 29% of the variation in public alignment. Adding firm fixed effects produces a larger increase, raising the explained share to 50.6%. This heterogeneity is consistent with our framework's premise that alignment is shaped by a firm-level characteristic, namely, the cost of providing political support. Second, we find that alignment rises substantially from 2003 to 2021, with the largest increases concentrated in the upper tail of the firm-year distribution.

Using this measure, we evaluate the empirical implications of the noise, government consumption, and screening explanations. Throughout, we account for other observable forms of political embeddedness and exposure that may be correlated with alignment, including Party cells,[3] state ownership, political connections, and political risk.

Our first empirical exercise tests the payoff-response restriction. Absent screening, aligned and nonaligned firms have the same mean payoff response when political stress materializes. We test this prediction using the July 2017 downfall of Sun Zhengcai, an incumbent Politburo member whose sudden removal was widely interpreted as evidence of elite instability within Party leadership rather than news about economic conditions. The sharp timing of the episode allows us to conduct a high-frequency event study using weekly stock returns of listed firms in our sample, comparing the responses of more- and less-aligned firms.

We find that more-aligned firms experienced significantly larger stock price declines during and

[3] Party cells are formal CCP organizations established within firms. They typically include firm employees who are Party members and provide an institutional channel through which the Party can disseminate directives, monitor political compliance, and influence corporate decision-making.

immediately after the scandal. Cumulating the effect over the announcement week and the two weeks that follow, the implied difference between firms at the 75th and 25th percentiles of alignment is 1.3 percentage points, or approximately RMB 250 million (USD 37 million) for the average firm. Within our framework, this differential valuation response rules in a screening role. Its direction is also consistent with optimal screening, under which more-aligned firms provide greater support and experience larger payoff losses when political stress materializes. The result is robust to controlling for firms' political connections to Sun Zhengcai and other observable forms of political embeddedness.

Our second empirical exercise tests the support-response restriction. Absent screening, aligned and nonaligned firms have the same support response when the regime's need for support rises. We test this prediction using episodes of local labor unrest. Unrest is a source of political stress, and employment provides a clear but costly margin of support: by expanding employment, or reducing it less, firms can mitigate unrest (Mueller, 2025; Wen, 2025).

We find that, when local unrest in a city and year is high relative to the provincial average that year, more-aligned firms in that city maintain higher employment than otherwise comparable less-aligned firms. In a year of elevated unrest, a firm one standard deviation higher in alignment retains about 398 additional employees, or 6.4% of mean employment. Within our framework, this differential employment response rules in a screening role for public alignment. Its direction is also consistent with optimal screening, under which more-aligned firms bear greater support burdens when political conditions deteriorate. This response is robust to controls for state ownership, board political connections, Party-cell adoption, and firm-level political risk. We find no systematic differential responses in revenue, profitability, or labor productivity, suggesting the employment response is not due to differential business performance.

We then test several additional predictions from our theoretical framework. Firms increase public alignment after regulatory investigations, consistent with investigations lowering firms' opportunity costs of support and making alignment more attractive. More-aligned firms also have higher corporate social responsibility scores, suggesting that alignment is associated with costly activities that advance

regime priorities. Finally, higher alignment is associated with more government subsidies and more government procurement contracts, consistent with favorable treatment from the regime.

Our interpretation of the results is subject to two qualifications. First, public alignment need not be the only instrument through which the regime screens firms. Private communication, political connections, ownership ties, Party organizations, and other forms of embeddedness may also convey information about firms' willingness or ability to provide support. Our claim is not that public alignment is the only screening channel, but that it is a distinct and measurable one. Second, public alignment may affect audiences other than the regime. Such effects could sort firms according to their costs of providing support, in which case alignment has a screening role under our definition even if the sorting is not deliberately induced by the regime. Alternatively, they could affect firm values or behavior directly through channels outside the basic model. We discuss these possibilities in Section 7.

Our work contributes to a literature on the political economy of autocracy. This literature emphasizes the information, control, and commitment problems that shape political and economic exchange under nondemocratic regimes (North, 1971; North and Weingast, 1989; Shleifer, 1996; Shleifer and Vishny, 1998; Acemoglu and Robinson, 2005; Svolik, 2012; Egorov and Sonin, 2024; Yang, 2024).[4] We are the first to propose, model, and empirically test public firm speech as a screening device. By attaching support burdens and favorable treatment to alignment choices, the regime induces firms to self-select: only those that can provide political support at relatively low cost choose to align. We derive restrictions from a framework that nests several explanations and test them in the data. More-aligned firms lose more value after an adverse political shock and provide more costly support during local unrest. These patterns rule in a screening role.

Our paper also speaks to work on business-state relations in China. Recent studies document the

[4] Related work studies coalition formation and elite management (Acemoglu et al., 2008; Egorov and Sonin, 2011; Jia et al., 2015), propaganda and censorship (King et al., 2014; Yanagizawa-Drott, 2014; Guriev and Treisman, 2019; Rozenas and Stukal, 2019), and other institutions of autocratic control (Gehlbach et al., 2016; Martinez-Bravo et al., 2022; Mueller, 2025; Wen, 2025).

expanding institutional presence of the Party in firms and the broader rise of Party-state capitalism (Yan and Huang, 2017; Lin and Milhaupt, 2021; Pearson et al., 2021; Jordan, 2024). Others examine how Party leadership, political control, and political influence shape corporate governance, firm value, and firm behavior (Liu and Zhang, 2019; Zhang, 2019; Chen et al., 2023; Xie et al., 2022).[5] Our contribution is to show how public firm speech can screen firms according to their costs of providing support. This screening role is distinct from, but can coexist with, the value of political proximity and government consumption.[6]

We also contribute to the literature on corporate political speech.[7] We study public firm speech in an autocracy, with the regime as a central audience, and show that such speech can screen firms for costly support. This mechanism differs from favor exchange in media-capture settings, where favorable coverage is rewarded with benefits (Szeidl and Szucs, 2021; Durante et al., 2022), and from broader accounts of firms' political power through lobbying, campaign contributions, political connections, public advocacy, and other channels of influence (Bombardini and Trebbi, 2025). We bring the mechanism to the data by constructing a generalizable measure of public alignment based on regime-specific phrases in firms' annual reports.

Finally, our theoretical framework relates to classic work on signaling and screening, in which privately informed agents reveal their types through observable choices (Spence, 1973; Rothschild and Stiglitz, 1976). Here, the observable choice is public political speech and the firm's private type is its cost of providing support. While alignment is itself costless, the costly support burden the regime attaches to each alignment choice enables screening. The geometry of the screening problem also parallels arguments from information design, in which information structures create informative

[5] Related work in other settings studies the value of political connections and the broader political power of firms. See Bombardini and Trebbi (2025) on firms' political power, Fisman (2001) on Indonesia under Suharto, and Ferguson and Voth (2008) on Nazi Germany.

[6] Related work documents that alignment may intensify when political uncertainty rises (Jiang et al., 2023; Zhang et al., 2026).

[7] Work in democracies, especially the United States, studies corporate political speech with consumers, investors, employees, or other market participants as important audiences (Bhagwat et al., 2020; Barari, 2024; Conway and Boxell, 2026; Ottonello et al., 2024; Cassidy and Kempf, 2025). Related work studies rhetorical nationalism among Chinese firms (Yue et al., 2024).

partitions of an underlying state space (Kamenica and Gentzkow, 2011; Bergemann and Morris, 2019). We apply this logic to public political speech and derive testable restrictions that distinguish screening from payoff-irrelevant speech and government consumption.

2 Background

We now motivate three potential explanations of public alignment: noise, government consumption, and screening. Historical and contemporary examples are consistent with each of the three explanations but do not distinguish among them. Section 7 considers explanations in which investors, consumers, employees, or other stakeholders are relevant audiences alongside the regime.

Alignment as noise. One possibility is that alignment is largely ceremonial. When aligned speech is nearly universal, it conveys little firm-specific information and may be payoff-irrelevant within the firm–regime relationship. Official slogans, for example, became a routine part of industrial life in the Soviet Union (Jacobs, 1957). In North Korea, propaganda slogans and posters similarly appear in factories and other public spaces.[8]

Alignment as government consumption. A second possibility is that the regime directly values both aligned speech and costly actions that advance its political objectives. Aligned speech may confer public legitimacy or satisfy the regime's demand for praise, while costly actions provide material support. The regime may reward firms that provide these benefits with contracts, licenses, subsidies, regulatory leniency, protection, or other favorable treatment. Such rewards give firms that do business with the regime a reason to adopt pro-government language as part of an ongoing commercial relationship.

Several examples are consistent with this interpretation. Multinational firms have adopted Beijing's preferred language on political issues to preserve access to the Chinese market.[9] In Russia,

[8] See NK News, 2021.
[9] See ChinaFile, 2018.

domestic companies have publicly appealed to Vladimir Putin for protection.[10] In Turkey and Hungary, pro-government editorial alignment has similarly been linked to state advertising and other official support.[11]

Alignment as screening. A third possibility is that alignment helps the regime screen firms according to their costs of providing support. Firms privately observe their costs of reallocating resources toward regime priorities, maintaining employment during unrest, or undertaking politically valuable investments. By linking alignment to support burdens and favorable treatment, the regime can induce lower-cost firms to align. Under optimal screening, public alignment is therefore informative because it predicts which firms take costly actions when political conditions deteriorate.

Several examples link public alignment with costly action. In China, the regime's "common prosperity" campaign called on firms to support a politically salient redistribution agenda after a period of heightened regulatory pressure on large private firms. Alibaba pledged RMB 100 billion (USD 15.5 billion) to common-prosperity initiatives, and other private firms announced similar pledges.[12] These announcements combined public endorsement of a regime priority with commitments of firm resources. In Russia, the founder of the shoe company Zenden publicly backed the war in Ukraine and was among the first Russian businessmen to invest in occupied Crimea after the 2014 annexation, pairing public loyalty with a costly and politically sensitive investment.[13] Evidence from Syria also describes regime-linked business networks whose access to public resources depends on the economic assistance they provide to the regime.[14] These examples do not establish screening but show that public alignment can be embedded in relationships that pair words with costly actions, in exchange for favorable treatment.

[10] See Financial Times, 2025.
[11] See Reuters Institute, 2017; Financial Times, 2025.
[12] See Wall Street Journal, 2021.
[13] See Meduza, 2026.
[14] See Brookings, 2024.

3 Framework

We now develop a unified framework that rationalizes alignment in the firm–regime relationship as noise, government consumption, or screening. We first introduce the basic model, then provide a microfoundation connecting it to our empirical context, and finally discuss modeling assumptions.

Basic model. We study a static interaction between a firm and a regime.[15] The firm privately observes its cost type $c \in [\underline{c}, \overline{c}] \subset \mathbf{R}_+$, distributed according to an absolutely continuous distribution F with positive density f. The type c represents the firm's opportunity cost of providing support to the regime. The regime knows F but does not observe the realization of c.

The interpretation of c is broad. Providing support may require the firm to direct labor, capital, managerial attention, or organizational capacity toward political rather than commercial objectives. During periods of labor unrest, for example, the regime may value expanding hiring, maintaining employment despite weak demand, or absorbing workers in politically sensitive regions. By raising the opportunity cost of participating in unrest (Becker, 1968), these actions can reduce social instability and help the regime maintain political order. They are costly to the firm because the resources involved could otherwise be used for production, investment, or restructuring.

Heterogeneity in c captures differences in opportunity costs of providing support. Cash contributions to a regime-prioritized campaign, for example, are more costly for a firm with valuable investment opportunities. Similarly, additional hiring during unrest is more costly for a cash-constrained firm that has difficulty financing a larger payroll. Thus, c summarizes the firm's cost of undertaking actions that advance the regime's sociopolitical objectives.

The regime can commit *ex ante* to a menu of contracts indexed by the firm's alignment choice $a \in \{0, 1\}$.[16] We interpret $a = 1$ as alignment and $a = 0$ as nonalignment.[17]

[15] In Appendix B, we show how our static model can be interpreted as a reduced-form representation of incentives sustained in an infinite-horizon repeated interaction.

[16] We focus on the case of binary alignment levels for simplicity, but the analysis can be extended to accommodate any finite number of alignment levels.

[17] The framework does not require alignment to take the form of speech that is publicly visible to investors, employees, or

For each alignment choice a, a contract (s_a, t_a) specifies a support burden $s_a \in [0, q]$, where $q \in (0, 1)$ is the maximum feasible burden, and an expected transfer payment $t_a \in \mathbf{R}$. The support burden represents the probability or intensity with which the firm provides costly support, as explained below. A positive transfer may represent a subsidy, government contract, regulatory leniency, or another benefit provided by the regime. A negative transfer may represent a tax, compliance cost, or another costly concession imposed on the firm.

Given the menu, the firm chooses $a \in \{0, 1\}$. A type-c firm that selects contract (s_a, t_a) receives expected payoff

$$t_a - cs_a,$$

while the regime receives

$$U_a + V_a s_a - t_a.$$

The term U_a captures the regime's direct value from the firm's alignment choice, including the value of public affiliation, symbolic loyalty, or the legitimacy conferred by visible endorsement. The term $V_a s_a$ captures the value of costly firm support. Allowing V_a to depend on alignment accommodates complementarities between alignment and support. For example, additional hiring during labor unrest may be more valuable when undertaken by a publicly aligned firm because the employment response is then more readily associated with the regime.

The alignment choice a is itself costless to the firm. It enters the firm's payoff only through the contract (s_a, t_a) it selects, so every type can align. Differences in firms' incentives to align therefore arise through the opportunity cost c of the support burden attached to each contract. When the menu attaches different support burdens to the two choices, lower-cost firms are more willing to select the choice with the greater burden. Under the surplus-maximizing separating menu characterized below, separation therefore runs through the support burdens attached to the two choices.

other outside audiences. It requires only that the firm's alignment choice—and, in the repeated-game interpretation, its history—be commonly observed by the firm and the regime. We discuss why alignment may nevertheless be publicly visible in Section 7.

We focus on menus that maximize expected total surplus. Assigning a type-c firm to contract (s_a, t_a) generates total surplus

$$U_a + V_a s_a - c s_a.$$

Thus, for a given alignment choice and support burden, the transfer affects how surplus is divided but not how much is generated. In Appendix B, we show how this separation can arise in a repeated interaction.

Political shocks and support. The support burden s_a summarizes the costly support that the firm expects to provide. Support need not follow an explicit regime demand; it may also reflect informal expectations within the firm–regime relationship or voluntary actions that advance regime objectives.

To interpret this burden, suppose the political state is $\theta \in \{B, G\}$, where G represents normal times and B represents a period of political stress. The bad state may represent labor unrest, social instability, geopolitical conflict, or another shock that increases the value of firm support to the regime. For simplicity, support is relevant only in the bad state, which is realized after the firm has chosen its alignment level. The state is drawn independently of the firm's type and alignment choice, with $\Pr[\theta = B] = q \in (0, 1)$. Let $x_a \in [0, 1]$ denote the probability or intensity with which a firm choosing alignment a provides support when $\theta = B$. The firm's *ex ante* support burden is therefore

$$s_a = q x_a \in [0, q].$$

We use this reduced-form burden in the basic model because it directly determines the firm's expected support cost.

This microfoundation captures the idea that support is most relevant during political stress. In our running example, the bad state is a period of labor unrest, during which the regime places greater value on employment stabilization. A firm can provide support by expanding employment or reducing it less than it otherwise would. The term x_a captures the firm's expected employment response conditional on unrest, while s_a captures its expected support burden before the political state is realized.

The linear payoff specification in the basic model can be derived from this microfoundation. Let $X \in \{0, 1\}$ denote realized support, with $\Pr[X = 1 \mid a, \theta = B] = x_a$. Suppose providing support X costs a type-c firm cX. Its expected payoff under contract (s_a, t_a) is then

$$t_a - qcx_a = t_a - cs_a,$$

as in the basic model. Now suppose the regime's gross state-contingent payoff is given by

$$v(X, a, \theta) = u_a(\theta) + V_a X \mathbf{1}\{\theta = B\},$$

where $u_a(\theta)$ is its direct value from alignment choice a in state θ and V_a is its marginal value of support in the bad state. Its expected gross payoff is

$$q\left[u_a(B) + V_a x_a\right] + (1 - q)\, u_a(G) = \underbrace{q u_a(B) + (1 - q)\, u_a(G)}_{=: U_a} + V_a s_a.$$

This microfoundation connects our theoretical and empirical approaches. Theoretically, we proceed with the reduced-form support burden s_a, but this object can be linked directly to political shocks. Empirically, we interpret political shocks as events that change the value of firm support to the regime, allowing us to study whether alignment predicts differential exposure to support burdens.

Discussion of modeling assumptions. We conclude this section by discussing three modeling assumptions: commitment, private information, and surplus maximization.

Commitment. Our basic model assumes that the regime can commit *ex ante* to the menu of contracts. This assumption is substantive. Autocratic regimes often face weak institutional constraints, making promises to private actors difficult to enforce (North and Weingast, 1989; Gehlbach and Keefer, 2011; Egorov and Sonin, 2024). In our setting, the regime may promise transfers, regulatory leniency, or limited support burdens but later have an incentive to renege.

We interpret commitment as a reduced-form representation of a longer-run firm–regime relationship. Repeated interaction can allow the parties to use current transfers and future treatment to sustain

arrangements that cannot be enforced formally (Levin, 2003). In Appendix B, we construct such a repeated interaction and derive conditions under which its effective transfers and stationary allocation coincide with those in the basic model. Formal commitment is therefore not necessary, although the relationship must generate sufficient continuation value to satisfy the relevant enforcement constraints.

Private information. A key friction in our model is that the firm privately observes its cost of support. Firms generally have better information than the regime about their productivity, outside opportunities, internal constraints, and the opportunity cost of redirecting resources toward political objectives.

Private information creates a potential screening role for alignment. If the regime observed each firm's cost, it could condition support burdens directly on that cost. When costs are private, the regime can instead attach different support burdens to different alignment choices. Because a lower-cost firm is more willing to accept a greater burden, such a menu can induce alignment choices that reveal firms' support costs. More generally, as we formalize in Section 4, alignment has a screening role whenever observing it changes the regime's posterior mean of c. The induced separation does not require alignment itself to be costly; it requires only that the attached support burden be more costly for some types than for others.

This friction is natural in our empirical context. Information about firms' opportunity costs is decentralized and difficult for the state to observe directly (Hayek, 1945). The issue is especially salient in China, where the state has extensive administrative capacity but continues to face informational constraints in governing local officials and other economic actors (Huang et al., 2017; Yang, 2024).

Surplus maximization. Our analysis focuses on menus that maximize expected total surplus. This identifies the efficient allocation of alignment and support. By contrast, transfers divide the resulting surplus between the firm and the regime without changing how much a given allocation generates. Surplus maximization does not require the firm and the regime to have equal bargaining power: bargaining power affects the division of surplus rather than the efficient allocation.

The repeated-interaction analysis in Appendix B formalizes this separation. When continuation

value is constant across alignment choices, current transfers and the division of continuation value can sustain the effective transfers in the basic model without altering allocative surplus.

4 Theoretical Analysis

We analyze the basic model in two steps. First, we derive a restriction that allows us to rule in a screening role for alignment: absent screening, alignment cannot predict differential mean payoff responses or support responses when the bad state is realized. Second, we characterize the surplus-maximizing menu and derive comparative statics on the value and prevalence of screening. Proofs are deferred to Appendix B.

4.1 Ruling In a Screening Role for Alignment

Consider a menu of contracts $\{(s_0, t_0), (s_1, t_1)\}$. Since the firm chooses the contract that maximizes its payoff, its alignment choice satisfies

$$a(c) \in \arg\max_{a \in \{0,1\}} \{t_a - c s_a\}.$$

If both alignment choices occur with positive probability, we say that *alignment has a screening role* if

$$\mathbf{E}[c \mid a = 1] \neq \mathbf{E}[c \mid a = 0].$$

Thus, alignment has a screening role when it changes the regime's posterior mean of the firm's support cost. If only one alignment choice occurs with positive probability, alignment has no screening role.

Let $s(c) = s_{a(c)}$ denote the support burden induced by the firm's alignment choice. Total surplus can be decomposed into three components:

$$\begin{aligned} W &= \int_{\underline{c}}^{\overline{c}} \left[U_{a(c)} + V_{a(c)} s(c) - c s(c)\right] \mathrm{d}F(c) \\ &= \underbrace{U_0}_{\text{noise}} + \underbrace{\int_{\underline{c}}^{\overline{c}} \left[U_{a(c)} - U_0 + \left(V_{a(c)} - \mathbf{E}[c]\right) s(c)\right] \mathrm{d}F(c)}_{\text{government consumption}} + \underbrace{\mathbf{Cov}[c, -s(c)]}_{\text{screening}}. \end{aligned}$$

This decomposition maps into the three explanations of alignment. First, the noise component is the value of nonalignment with no support. Second, the government consumption component captures the regime's value from the chosen alignment and support burden, evaluated at the average cost of support. It therefore includes both the direct value of alignment and the direct value of costly firm support. Third, the screening component captures the gain from assigning greater support burdens to lower-cost firms. Because incentive compatibility makes $s(c)$ nonincreasing in c, this component is nonnegative. Moreover, observe that

$$\mathbf{Cov}[c, -s(c)] = (s_1 - s_0)\Pr[a = 1]\Pr[a = 0]\left[\mathbf{E}[c \mid a = 0] - \mathbf{E}[c \mid a = 1]\right].$$

Consequently, when both alignment choices occur with positive probability and $s_1 \neq s_0$, the screening component is positive when alignment has a screening role.

Proposition 1. *Consider any menu* $\{(s_0, t_0), (s_1, t_1)\}$ *under which both alignment choices occur with positive probability. If alignment has no screening role, then* $s_1 = s_0$ *and* $t_1 = t_0$.

Proposition 1 provides the first step in our analysis. It applies to any menu and does not assume surplus maximization. Since $s_a = qx_a$, equal support burdens imply $x_1 = x_0$. If alignment has no screening role, the two groups also have the same mean support cost. The realization of a bad state therefore produces the same mean payoff response and the same support response in both groups. Observing a differential response on either margin rules in a screening role within the framework.

4.2 Optimal Screening

Having established a restriction implied by the absence of screening, we now characterize the surplus-maximizing menu. To focus on the case consistent with our empirical evidence, we impose the following condition on how the regime values alignment and support.[18]

[18] Without this assumption, the surplus-maximizing menu may instead assign a greater support burden to nonaligned firms. The model can also be solved in this case, but this reverse orientation does not match the empirical patterns we document below.

Assumption 1. *The regime's value from alignment and support satisfies*

$$U_1 + qV_1 \geq U_0 + qV_0 \qquad \textit{and} \qquad V_1 \geq \overline{c}.$$

The first inequality states that, at the maximal support burden, alignment is at least as valuable to the regime as nonalignment. The second states that the regime's marginal value of support from an aligned firm is at least as large as the highest possible cost of providing it. Thus, conditional on alignment, maximal support is surplus maximizing for every firm type. Together, the two inequalities imply that alignment with maximal support generates at least as much surplus as either alignment without support or nonalignment with maximal support.

Proposition 2. *Under Assumption 1, there exists a surplus-maximizing menu* $\left\{\left(s_0^*, t_0^*\right), \left(s_1^*, t_1^*\right)\right\}$ *with* $s_0^* = 0$ *and* $s_1^* = q$*. Under this menu, the firm aligns if and only if*

$$c \leq c^* := V_1 + \frac{U_1 - U_0}{q}.$$

Moreover, if both contracts are chosen by a positive measure of types, then

$$t_1^* - t_0^* = qc^* = U_1 - U_0 + qV_1.$$

Proposition 2 provides the second step in our analysis. When $c^* \in (\underline{c}, \overline{c})$, the surplus-maximizing menu separates firm types: lower-cost firms align, bear the maximal support burden, and receive a higher transfer. In particular, $\mathbf{E}[c \mid a = 1] < c^* < \mathbf{E}[c \mid a = 0]$, so alignment has a screening role. Because $s_1^* = q > s_0^* = 0$, the menu also uses this information to assign the greater support burden to the lower-cost group. Equivalently, $x_1^* = 1 > x_0^* = 0$. When the bad state is realized, an aligned firm provides support and its expected support cost rises by $(1 - q)\, c$ relative to its *ex ante* expected support cost, while a nonaligned firm provides no support and has no corresponding payoff response.

However, the surplus-maximizing menu need not separate firms. If $c^* \notin (\underline{c}, \overline{c})$, all types choose the same contract and screening through distinct contracts generates no surplus gain. This raises a natural question: when does alignment have value as a screening device?

To answer this question, consider the maximal surplus generated by assigning type c either nonalignment without support or alignment with maximal support:

$$\Phi(c) \coloneqq \max\{U_0, U_1 + (V_1 - c)\, q\}.$$

By Proposition 2, the optimal two-contract menu generates expected surplus $\mathbf{E}[\Phi(c)]$. If the regime is restricted to a single contract, it cannot screen firms and must choose the same alignment and support burden for every type. Under Assumption 1, its maximal expected surplus is $\Phi(\mathbf{E}[c])$. We define the value of screening as

$$\Sigma \coloneqq \mathbf{E}[\Phi(c)] - \Phi(\mathbf{E}[c]).$$

This is generally distinct from the screening component $\mathbf{Cov}[c, -s(c)]$: it measures the gain from the optimal two-contract menu relative to the optimal pooling contract.

The value of screening has an intuitive geometric interpretation, shown in Figure 1. The function Φ is the upper envelope of the surplus generated by nonalignment without support (in red) and alignment with maximal support (in green). Both contracts generate the same surplus at c^*.

When $c^* \in (\underline{c}, \overline{c})$, the menu splits firms into an aligned group with average cost $\mathbf{E}[c \mid a = 1]$ and a nonaligned group with average cost $\mathbf{E}[c \mid a = 0]$. Since

$$\mathbf{E}[c] = \Pr[a = 1] \cdot \mathbf{E}[c \mid a = 1] + \Pr[a = 0] \cdot \mathbf{E}[c \mid a = 0],$$

the prior mean is a convex combination of the two group means. Moreover, Φ is affine within each group. It follows that

$$\mathbf{E}[\Phi(c)] = \Pr[a = 1] \cdot \Phi(\mathbf{E}[c \mid a = 1]) + \Pr[a = 0] \cdot \Phi(\mathbf{E}[c \mid a = 0]).$$

Thus, $\mathbf{E}[\Phi(c)]$ is the value at $\mathbf{E}[c]$ of the chord joining the two group means, while $\Phi(\mathbf{E}[c])$ is the value without splitting. The value of screening is the vertical distance between this chord and the upper envelope. This geometry is closely related to similar arguments in information design.

This geometric interpretation yields an immediate characterization of when optimal alignment has

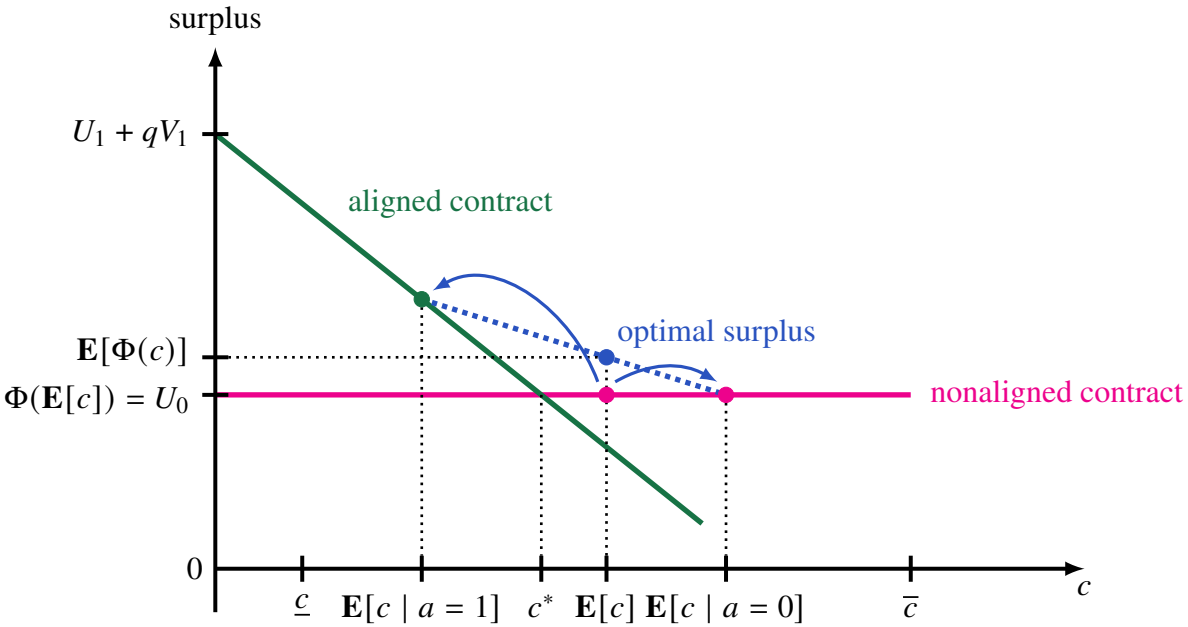


Figure 1: The value of screening when $c^* < \mathbf{E}[c]$; the case $c^* > \mathbf{E}[c]$ is analogous.

screening value and how it changes with our model primitives.

Proposition 3. *Under Assumption 1 and given the surplus-maximizing menu characterized in Proposition 2, the value of screening Σ satisfies the following properties.*

(i) $\Sigma > 0$ *if and only if* $c^* \in (\underline{c}, \overline{c})$.

(ii) Σ *is nondecreasing in* c^* *when* $c^* < \mathbf{E}[c]$ *and nonincreasing in* c^* *when* $c^* > \mathbf{E}[c]$.

(iii) *Among distributions supported on* $[\underline{c}, \overline{c}]$, Σ *is nondecreasing in mean-preserving spreads of* F.

Proposition 3 identifies the environments in which alignment is most valuable as a screening device. Consistent with the geometric intuition in Figure 1, screening has the highest value when the cutoff coincides with the average opportunity cost of support $\mathbf{E}[c]$. Greater dispersion in firms' support costs also weakly increases the value of screening.

4.3 Empirical Implications

While the value of screening is difficult to measure empirically, the comparative statics characterized in Proposition 3 have straightforward implications for the probability of alignment.

Proposition 4. *Under Assumption 1, the* ex ante *probability that the firm aligns is nonincreasing under first-order stochastic dominance shifts of F and nondecreasing in c^*.*

As Proposition 4 shows, the probability of alignment and the value of screening need not move together. The probability of alignment is nondecreasing in c^*, whereas the value of screening is nondecreasing when $c^* < \mathbf{E}[c]$ and nonincreasing when $c^* > \mathbf{E}[c]$. Alignment can therefore become more prevalent even as the value of screening declines, eventually reaching zero when all firms align.

Taken together, the results in this section provide two sets of empirical implications:

1. **Testing for the screening role of alignment.** Proposition 1 implies that, absent screening, mean payoff responses and support responses cannot differ by alignment when the bad state is realized. Observing either differential response rules in a screening role within our framework.
2. **Interpreting observed patterns of alignment.** Propositions 2 to 4 characterize how alignment is optimally used as a screening device and how its prevalence varies with primitives. Table A.2 summarizes the resulting mapping from primitives to predicted empirical patterns. These results provide a framework for interpreting observed patterns of alignment and their correlates through the lens of optimal screening.

5 Measuring Public Alignment

Our theoretical results generate predictions about which firms align, how the regime treats them, and how their mean payoffs and support behavior respond to political stress. Testing these predictions requires a firm-level measure of public alignment that is transparent, regime-specific, and reproducible from public sources. We construct such a measure from listed firms' annual reports. These reports are public, systematically archived, and written for external audiences, including investors, regulators, and other stakeholders. They therefore provide a consistent source of firm speech across firms and over time. Appendix A.2 documents that Chinese government agencies systematically read these filings and, separately, publicize and reward visible political alignment among listed firms.

Our measure draws on two text corpora: official CCP publications and the annual reports of listed Chinese firms. We first use the regime corpus to identify 50 phrases that are common in official Party discourse but uncommon in standard Mandarin. We then record whether firms use these phrases in their annual reports and aggregate the resulting indicators to the firm-year level using their first principal component. The resulting index assigns higher values to firms whose reports use more regime-specific language.[19]

China is a useful setting for our empirical exercise because it is the world's largest autocracy by GDP, firms' use of Party language is widespread, and large corpora of comparable regime and firm texts are available over two decades. The measurement procedure itself is general and can be applied to other settings with comparable corpora of regime and firm speech.

Regime corpus. The corpus of CCP publications contains three types of documents: *(i)* State Council and General Office circulars announcing policies and implementation guidelines; *(ii)* Central Committee plenary reports that record meeting minutes from major CCP meetings; and *(iii)* articles from the *People's Daily*, the official newspaper of the CCP. The combined corpus spans 1973–2023, with the densest coverage after the early 2000s. It contains 248 circulars, 36 reports from plenary sessions, and 617,937 *People's Daily* articles.[20] Because the firm corpus begins in 2003, our main firm-year analyses use annual reports from 2003 onward.

Phrase scoring and normalization. Within each document, we use the `jieba` dictionary to segment Chinese text into phrases, or multi-character tokens.[21] For a phrase i in document j of type t, we compute a term-frequency-inverse-document-frequency (TF-IDF) score:

$$\text{tfidf}_{ijt} = \text{tf}(i,j) \times \left[\log\left(\frac{1 + n_t}{1 + \text{df}(i)_t} \right) + 1 \right],$$

[19] In robustness checks, we show that this measure is substantially correlated with a more flexible LLM-based measure, while remaining more transparent and easier to interpret (see Appendix C.1).

[20] Our baseline dictionary uses all three types of documents. We weight sources equally when constructing phrase scores so that sources with more documents do not mechanically receive greater weight.

[21] We use the default `jieba` dictionary for both segmentation and Chinese language usage frequency. For information on the `jieba` dictionary, see https://github.com/fxsjy/jieba.

where $\mathrm{tf}(i, j)$ is the within-document frequency of phrase i in document j, n_t is the number of documents of type t, and $\mathrm{df}(i)_t$ is the number of type-t documents containing phrase i. Each document's vector of TF-IDF scores is then scaled to unit Euclidean (L^2) length, the default normalization in scikit-learn, the package we use to compute these scores. We then average across documents and document types so that no single source dominates because of its volume:

$$\text{Corpus tfidf}_i = \frac{1}{T}\sum_{t=1}^{T}\frac{1}{J_t}\sum_{j=1}^{J_t}\text{tfidf}_{ijt}, \qquad T = 3,$$

where T is the total number of document types and J_t is the number of documents of type t.

To identify phrases distinctive to Party discourse relative to standard Mandarin, we divide the `jieba` baseline word-frequency vector by its own Euclidean norm, so that both scores are dimensionless, and subtract the phrase's baseline frequency in the default `jieba` dictionary:

$$\text{Final tfidf}_i = \text{Corpus tfidf}_i - \text{Mandarin Frequency}_i.$$

A high Final tfidf_i score therefore indicates that a phrase is disproportionately common in regime publications relative to standard Mandarin usage.

Dictionary construction. We rank candidate phrases by Final tfidf_i and remove tokens of no more than two characters because one- and two-character Chinese tokens are often generic or ambiguous and therefore provide less specific indicators of regime language. Our baseline Party dictionary consists of the 50 highest-ranked remaining phrases, listed in Table 1. These include names of senior leaders, Party institutions, and hallmark policies. Some phrases can also appear in nonpolitical contexts despite being disproportionately common in Party discourse.[22]

Firm corpus. We obtain the annual reports of Chinese A-share firms from WIND.[23] These reports are standardized by securities regulation and contain narrative discussions alongside financial statements. In 2021, the total operating revenue of Chinese A-share firms was equivalent to approximately 47.8%

[22] Our primary aggregation procedure, described later in this section, accounts for this possibility by placing greater weight on phrases that tend to appear together and less weight on phrases used idiosyncratically.

[23] See https://www.wind.com.cn/mobile/EDB/en.html for details.

Table 1: Machine-Selected Phrases

Phrase	**English Translation**	**TF-IDF Diff** $\times 10^3$	**Phrase**	**English Translation**	**TF-IDF Diff** $\times 10^3$
贯彻落实	ensure effective implementation	9.678	建立健全	establish and improve	6.526
习近平	Xi Jinping	8.994	各族人民	people of all ethnic groups	6.470
重要讲话	important speech	8.164	党的建设	Party building	6.341
总书记	general secretary	8.100	主持会议	chair a meeting	6.266
党中央	Party Central Committee	8.016	充分肯定	fully affirm	6.062
深化改革	deepen reform	7.466	宏观调控	macroeconomic regulation	5.987
发展观	outlook on development	7.444	中共中央	CCP Central Committee	5.980
全体会议	plenary session	7.421	相结合	integrate	5.944
充分发挥	fully leverage the role of	7.294	管理体制	governance system	5.886
三个代表	Three Represents	7.281	伟大旗帜	great banner	5.814
改革开放	Reform and Opening Up	7.274	全党全国	the whole Party and the nation	5.797
负责同志	responsible officials	7.204	社会保障	social security	5.780
中央委员	Central Committee member	7.201	党的领导	Party leadership	5.671
中央纪律检查委员会	Central Commission for Discipline Inspection	7.144	讨论稿	draft for discussion	5.598
体制改革	institutional reform	7.068	公共服务	public services	5.583
中央政治局	Politburo	7.043	中央书记处	Central Secretariat	5.557
中央委员会	Central Committee	6.948	中央军事委员会	Central Military Commission	5.504
经济社会	economy and society	6.886	认真贯彻	rigorously implement	5.470
列席会议	attend the meeting	6.875	积极性	initiative	5.467
邓小平理论	Deng Xiaoping Theory	6.862	直属机构	directly affiliated institutions	5.413
党和国家	the Party and the state	6.816	各部委	ministries and commissions	5.412
常务委员会	standing committee	6.810	姓氏笔画	surname stroke count	5.363
小康社会	moderately prosperous society	6.807	坚定不移	unswerving	5.293
进一步	further	6.740	现代化	modernization	5.269
胡锦涛	Hu Jintao	6.561	各项事业	various undertakings	5.263

Notes: The table lists the 50 highest-ranked phrases after excluding tokens of no more than two characters. Scores are constructed by averaging TF-IDF across three types of CCP documents and subtracting its normalized frequency in standard Mandarin. Higher scores indicate phrases that are common in CCP documents relative to standard Mandarin.

of China's GDP. Our corpus contains 49,282 reports from 4,096 firms. Appendix A.1 describes the institutional process through which annual reports are produced in China, and Appendix D presents further details on the firm corpus.

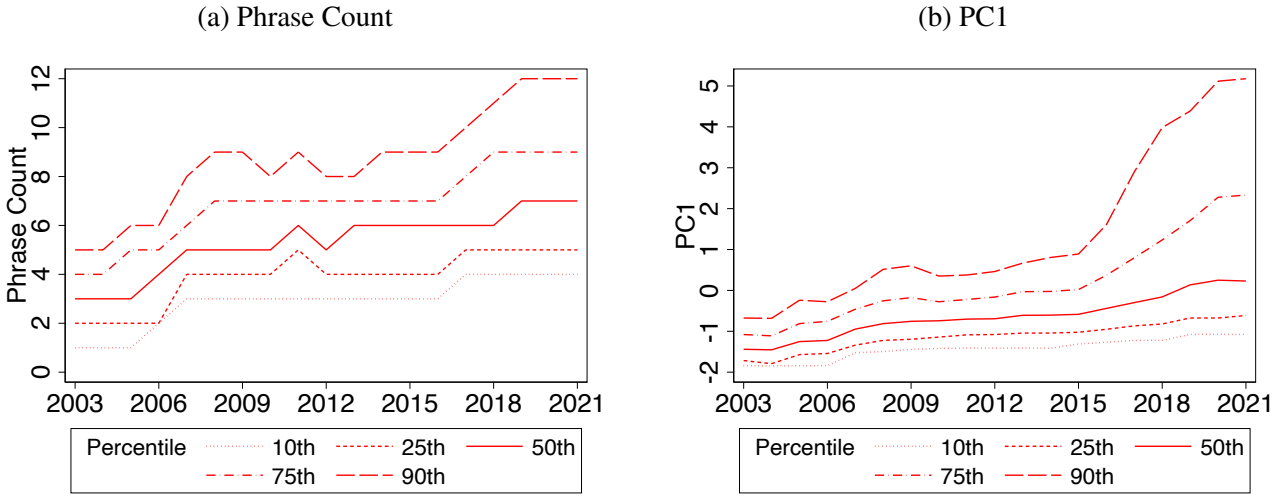
Public alignment. For each firm-year, we construct 50 indicators, each equal to one if a Party-dictionary phrase appears in any annual report released by that firm during that calendar year. Because the dictionary comprises phrases that are disproportionately common in official Party documents and relatively rare in standard Mandarin, these indicators capture language characteristic of CCP rhetoric rather than generic references to politics, policy, or macroeconomic conditions. Contextual validation, described in Appendix C.1, shows that the surrounding text is positive and that firms use these phrases affirmatively, rather than in a neutral or critical tone. As we detail in Appendix C.2, we therefore interpret the measure as capturing public alignment with the CCP, distinct from general discussion of political topics or exposure to political risk, as measured by Hassan et al. (2019).

We construct two measures from these indicators. The first, *Phrase Count*, is the number of distinct Party phrases, out of 50, used by the firm that year. It weights each phrase equally and is directly interpretable. The second, *PC1*, is the first principal component of the 50 indicators. It places greater weight on phrases that tend to appear together across firm-years and less weight on phrases whose occurrence in isolation may reflect incidental, nonpolitical usage. PC1 explains 7.12% of the total variance across phrases. We use the PC1 score as our primary alignment index; in all regressions, the score is standardized within the estimation sample.[24]

Public alignment increases over time. Figure 2(a) plots the distribution of Phrase Count. The median rises from 3 phrases in 2003 to 7 in 2021, while the interquartile range expands from $[2, 4]$ to $[5, 9]$ and the 90th percentile rises from 5 to 12. Figure 2(b) shows a similar pattern for PC1. Alignment rises gradually through the 2000s, levels off around 2011–2014, and increases more rapidly after 2016. The increase is most pronounced in the upper tail, while many firms continue to exhibit relatively little

[24] The first principal component loads positively on 46 of the 50 phrases. The four negative loadings are small in magnitude and correspond to relatively apolitical phrases: "attend the meeting" (-0.0055), "host a meeting" (-0.0137), "surname strokes" (-0.0043), and "macroeconomic control" (-0.0015).

Figure 2: Public Alignment Over Time



Notes: This figure plots public alignment among listed Chinese firms from 2003 to 2021. The unit of observation is a firm-year, where firm annual reports are obtained from the WIND Financial Terminal. The 50 Party-related phrases are identified using TF-IDF analysis on a corpus of CCP documents. In both panels, each line represents a different percentile of the firm-year distribution: the 10th (dotted), 25th (short-dashed), 50th (solid), 75th (dash-dot), and 90th (long-dashed) percentiles. Figure 2(a) displays the number of distinct Party-related phrases, out of 50, used in a firm's annual report. Figure 2(b) displays the first principal component of the 50 phrase indicators.

public alignment.

We note two key facts. First, alignment remains substantially dispersed among firms, even as its overall prevalence increases. Second, much of the increase occurs in the upper tail rather than uniformly across firms. This heterogeneity is consistent with our framework's premise that firms differ in their costs of providing support, creating scope for screening. However, it does not by itself distinguish among the three explanations. While we do not seek to explain the aggregate trend, the rise in alignment is consistent with our theoretical result that alignment becomes more prevalent as its value to the regime rises relative to firms' expected costs of support. In Section 6.3, we document additional firm-level correlates of alignment.[25]

We next assess how much variation in public alignment is explained by year, industry, city, industry-

[25] We also assess robustness to alternative dictionary and aggregation choices. Figures A.6(a) and A.6(b) restrict attention to the first 30,689 words of each annual report, equal to one standard deviation of average report length, to address the possibility that rising alignment mechanically reflects longer reports. Figure A.6(c) uses phrase counts per annual report instead of phrase indicators to capture intensity rather than presence. All three alternatives produce the same qualitative pattern: alignment rises over time, with the largest increases in the upper tail beginning in the mid-2010s.

by-city, industry-by-city-by-year, and firm fixed effects. As Table A.3 shows, year fixed effects alone yield an adjusted R^2 of 18.4%. Sequentially adding industry, city, and industry-by-city fixed effects raises the adjusted R^2 to 22.1%, 25.3%, and 29.4%, respectively, while adding industry-by-city-by-year fixed effects does not raise it further (28.7%; the slight decrease reflects the adjustment for added parameters). Adding firm fixed effects produces the largest increase, raising the adjusted R^2 to 50.6%.

These results show that public alignment varies substantially at both the firm and firm-year levels. The variation remaining after accounting for year fixed effects indicates that alignment is not merely a common time trend. City and industry fixed effects explain only modest additional shares, suggesting that alignment is not simply concentrated in more politically loyal places or particular sectors, such as banking or mining. Industry-by-city fixed effects likewise add relatively little, making it unlikely that alignment is driven primarily by norms or pressures within local markets. Allowing those local-market conditions to vary year by year changes little. Firm fixed effects explain a substantially larger share, indicating that persistent firm characteristics matter. Even so, roughly half of the variation remains at the firm-year level, suggesting an important role for within-firm changes in costs, constraints, or other conditions.

Other data. We combine the alignment measure with firm-level data on accounting, ownership, stock returns, government procurement, and regulatory investigations. Appendix D describes these sources in detail. Table A.4 provides an overview of the variables used in the analysis and their sources, and Table A.5 reports summary statistics.

6 Empirical Analysis

We now evaluate the empirical implications of our framework. Our first two exercises examine whether realizations of political stress affect the mean payoffs and support behavior of more- and less-aligned firms differently. Within our framework, a differential response on either margin rules in a screening role for alignment. The direction of these differences tests whether alignment operates as predicted

under optimal screening. We then examine additional implications of optimal screening for firms' alignment choices, costly activities, and treatment by the regime. Our theoretical predictions and corresponding empirical tests are summarized in Table A.2.

6.1 Firm Valuation After Adverse Political Events

Our first empirical exercise examines firm valuation after an adverse political event. Proposition 1 implies that, absent screening, public alignment cannot predict differences in firms' mean payoff responses. Proposition 2 predicts the direction of this response under optimal screening: more-aligned firms should experience larger valuation losses when the bad state is realized. We therefore ask whether an adverse political event affects the market values of more- and less-aligned firms differently and, if so, whether more-aligned firms experience larger losses.

The ideal event is an unexpected realization of the bad state that increases the value of firm support to the regime without directly affecting firm fundamentals, such as consumer demand or input costs. A scandal involving a senior CCP official can approximate such an event by creating a period of political stress within the leadership. In our framework, the scandal is a realization of the bad state: it increases the value of firm actions that preserve employment, investment, or economic confidence. At the same time, it need not directly affect firms' productivity or demand. Thus, we search for scandals involving incumbent Politburo members during our stock-return sample period and manually verify the timing and relevance of each case.[26]

We focus on the July 2017 announcement that Sun Zhengcai was under investigation. This was the only case involving an incumbent Politburo member in our sample period and the most prominent leadership scandal in the sample. It was also widely interpreted as a negative signal about elite political stability, rather than as news about economic conditions.[27] Sun was a former Chongqing Party Secretary, an incumbent Politburo member, and widely viewed as a rising figure in the CCP leadership.

[26] In Appendix E, we widen the search to include all officials at national deputy rank or above. We exclude military officials because military and civilian ranks are not directly comparable.

[27] Our empirical strategy is related to that of Liu et al. (2017), who study the effect of the Bo Xilai scandal on the stock returns of politically sensitive firms. The Bo scandal falls outside our sample period.

On July 24, Xinhua announced that Sun was under investigation for "serious violations of discipline," confirming reports about his political downfall and abruptly ending his career. The announcement was sudden from the perspective of outside observers and prompted widespread speculation about conflict within the senior Party leadership.[28]

We estimate a high-frequency event study using weekly stock returns around the announcement. The estimating equation is

$$\text{Return}_{i,t} = \sum_{k \in \{-2,0,1,2\}} \beta_k \left(\text{Alignment}_i \times \mathbf{1}\{t = k\} \right) + \zeta_i + \tau_t + \varepsilon_{i,t}. \tag{1}$$

The outcome is firm i's weekly stock return, and event week $t = 0$ is the week of the announcement. Alignment_i is measured using the firm's most recent annual report released before the event. The specification includes firm and event-week fixed effects, ζ_i and τ_t, and standard errors are clustered at the firm level. We restrict the sample to the five-week window from two weeks before to two weeks after the announcement and normalize the coefficient for week -1 to zero. Both stock returns and alignment are standardized within the event-window sample.

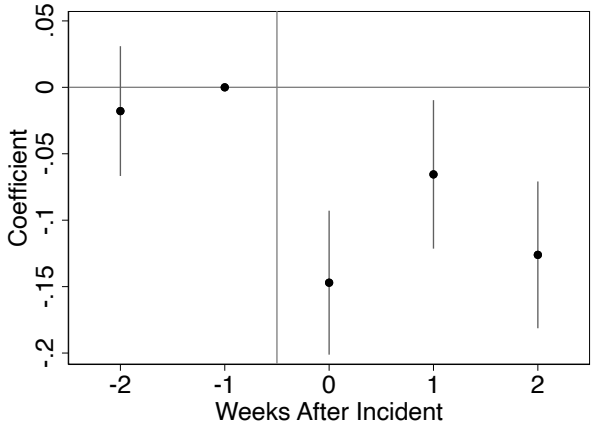
The event-study estimates show that more-aligned firms experienced lower stock returns during and immediately after the announcement, as reported in Figure 3(a). Within our framework, this differential valuation response rules in a screening role. Its direction is also the one predicted by optimal screening, under which more-aligned firms are more exposed when the bad state is realized. The same response could arise outside the basic model if alignment instead marked firms holding a larger stock of regime-linked rents that an unstable leadership devalues. We address that interpretation next.

The magnitude of this result is economically meaningful. Cumulating the estimates over the announcement week and the two weeks that follow, the implied difference between firms at the 75th and 25th percentiles of alignment is 1.3 percentage points. For a firm with average market capitalization

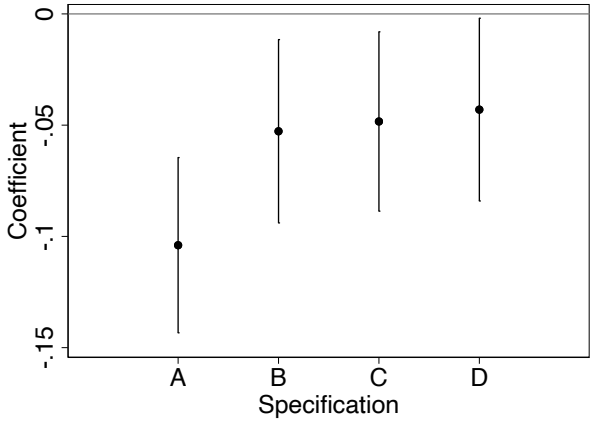
[28] See https://thediplomat.com/2017/07/chinese-politician-sun-zhengcai-is-under-party-investigation/.

Figure 3: Weekly Stock Returns Around the Sun Zhengcai Scandal

(a) Baseline



(b) Controlling for Closeness to the State



Notes: The unit of observation is a firm-week. Weekly stock returns are constructed from CSMAR prices as the percentage change from the opening price on the first trading day of the week to the closing price on the last trading day of the week. The sample is restricted to an event window from two weeks before to two weeks after the scandal. Both the outcome and the alignment measure are standardized within this window. Figure 3(a) plots coefficients on interactions between firm-level alignment and event-week indicators, with week -1 omitted as the reference period. The regression includes firm and event-week fixed effects. Figure 3(b) reports coefficients on $\text{Post}_t \times \text{Alignment}_i$, where Post_t equals one for the event week and subsequent weeks. Specification A includes firm and event-week fixed effects. Specification B additionally interacts Post_t with SOE status. Specification C additionally interacts Post_t with political connections. Specification D additionally interacts Post_t with Party-cell adoption. Standard errors are clustered at the firm level. Bars show 95% confidence intervals.

of RMB 18.6 billion, this amounts to approximately RMB 250 million (USD 37 million).

A potential concern is that public alignment captures other forms of closeness to the state that also expose firms to adverse political events (Fisman, 2001). We therefore allow the post-announcement response to vary with three standard measures of state closeness: state ownership, political connections, and Party-cell adoption.[29]

Figure 3(b) reports the coefficient on the interaction between alignment and the post-announcement indicator across four specifications. The first includes the baseline fixed effects. The remaining specifications sequentially add interactions between the post-announcement indicator and state ownership, political connections, and Party-cell adoption. The alignment coefficient remains negative throughout,

[29] State ownership is constructed from the Equity Nature data in CSMAR. Political connections are from Fan (2021) and indicate whether a firm had at least one politically connected director as of October 2013, before incumbent and recently retired government officials were barred from serving on the boards of listed firms. The Party-cell variable equals one from the first year in which the firm's corporate-charter amendment mentions establishing a Party cell onward.

although controlling for state ownership attenuates its magnitude. This attenuation suggests that state ownership captures some of firms' exposure to adverse political shocks. Nevertheless, public alignment captures additional variation not explained by these measures of political embeddedness. The differential return remains negative after controlling for state ownership, board political connections, and Party-cell adoption.

We conduct four further robustness checks. First, we re-estimate the event study using abnormal returns to account for differences in firms' exposure to broader market movements, as reported in Figure A.7(a).[30] Second, we control for whether the firm is headquartered in Sun's hometown, in a city where he was educated, or in a city where he previously held office, as reported in Figure A.7(b). The post-announcement alignment coefficient remains negative and statistically significant across these exercises, indicating that the result is not driven by firms tied to Sun through their location. Third, we control for a measure of political risk and find that the result survives, as shown in Appendix C.2. Fourth, we broaden the set of events to include scandals involving civilian officials at the national-deputy rank or above during our stock-return sample period. This search adds two events: the November 2021 public accusation against former Vice Premier Zhang Gaoli and the February 2018 sanctioning of former State Councilor Yang Jing for serious violations of Party discipline.[31] Figures A.7(c) and A.7(d) report the corresponding event studies. Both show negative event-period coefficients for more-aligned firms, but each also contains a statistically significant week -2 coefficient relative to the omitted week -1, with opposite signs across the two events. We therefore view these cases as suggestive evidence that the relationship extends beyond Sun Zhengcai, rather than as clean replications of the baseline event study.

[30] Appendix D.8 describes the construction of abnormal returns using standard event-study methods (MacKinlay, 1997).

[31] We define the scandal search space using the Chinese administrative hierarchy. We searched all civilian officials at the national deputy rank (*fuguoji*) or above during the period for which we observe stock returns, 2015–2022, excluding military officers because PLA ranks are not directly comparable to civilian administrative ranks. We included cases in which the official first became the subject of either a formal disciplinary action or a major allegation that generated widespread political attention during this period. Under this rule, the relevant cases are Sun Zhengcai, Zhang Gaoli, and Yang Jing. Sun Zhengcai is the only case involving an incumbent Politburo member in our sample period.

6.2 Costly Support During Political Crises

Our second empirical exercise turns from firm valuation to costly support. Since $s_a = qx_a$, Proposition 1 implies that, absent screening, public alignment cannot predict differences in firms' support responses when the bad state is realized. Proposition 2 predicts that, under optimal screening, more-aligned firms should provide greater support. We therefore test whether more publicly aligned firms undertake more costly actions that advance regime objectives during periods of political stress.

Local labor unrest provides a useful setting in which the bad state varies across cities and years. Social instability is politically salient in an autocracy, making unrest a natural realization of political stress.[32] Employment also provides a clear but costly margin of support. By expanding employment, or reducing it less, firms can help mitigate unrest. Higher employment raises potential participants' opportunity costs of engaging in unrest (Becker, 1968; Wen, 2025). Because unrest varies across cities and years, we can compare firms facing different local conditions at the same time. This exercise complements the scandal analysis, which studies a common high-frequency political event.

We estimate a distributed lead-and-lag specification relating firm employment to the interaction between public alignment and labor unrest in the firm's home city, using an empirical design related to Mueller (2025):

$$\begin{aligned}\text{Employees}_{i,t} = &\sum_{k=-2}^{2} \beta_k \left(\text{Alignment}_{i,t} \times \text{Unrest}_{c(i),t-k}\right) \\ &+ \delta \,\text{Alignment}_{i,t} + \sum_{k=-2}^{2} \gamma_k \,\text{Unrest}_{c(i),t-k} + \alpha_i + \gamma_{p(i),t} + \varepsilon_{i,t}.\end{aligned} \tag{2}$$

The outcome is the total number of employees reported by firm i in year t, drawn from the China Listed Firms' Corporate Governance Research Database in CSMAR; $c(i)$ denotes firm i's home city and $p(i)$, its province. We use employment levels because the regime's social-stability objective

[32] Labor unrest is common in China. Although the regime uses repression, surveillance, and public-security measures to address unrest, it does not suppress all local labor unrest. Local strikes can provide information to the central government about local grievances and officials, and suppressing all unrest would deprive the center of this information (Lorentzen, 2013).

depends on the absolute number of workers kept in employment, not the proportional change relative to firm size. A 1% change at a firm with 100,000 employees represents 1,000 jobs, whereas a 5% change at a firm with 200 employees represents only 10 jobs. Headcount is therefore the natural unit for the support burden, the magnitude calculation, and the comparison with the valuation response. The analysis is conducted at the firm-year level from 2003 to 2019. $\text{Alignment}_{i,t}$ is the firm-year public-alignment index constructed in Section 5.

The specification includes firm fixed effects α_i and province-year fixed effects $\gamma_{p(i),t}$. Firm fixed effects absorb time-invariant differences across firms, including persistent differences in size, political orientation, and average exposure to unrest. Province-year fixed effects absorb shocks and trends common to firms within the same province and year. We therefore leverage variation in unrest across cities within the same province and year. The specification also includes the uninteracted unrest variables at every lead and lag and the standalone alignment measure. The coefficients of interest are the β_k's, which trace whether more- and less-aligned firms respond differently before, during, and after local unrest. We hold the sample fixed across the full lead-lag specification and cluster standard errors at the city level.

Figure 4(a) presents the estimates.[33] The contemporaneous interaction between alignment and unrest is positive and statistically significant at $p < 0.05$, and it is the largest point estimate in the specification. More-aligned firms therefore expand employment, or reduce it less, in the year in which local unrest occurs. Both post-unrest estimates are indistinguishable from zero. We note that the pre-period estimates are individually significant: the estimate from two years before unrest is significant at $p < 0.05$, and the estimate from one year before unrest is significant at $p < 0.1$. However, if more-aligned firms in cities heading toward unrest had been on a separate, pre-existing employment path, one might expect that differential to widen as unrest approached. It narrows instead, from less than half the contemporaneous estimate two years before unrest to about a sixth of it one year before, before rising sharply in the unrest year. This pattern is inconsistent with a simple differential trend.

[33] Column (1) of Table A.6 reports the corresponding estimates in table form.

The magnitude is economically meaningful. A one-standard-deviation increase in local unrest corresponds to approximately 20 additional strikes. During such a year, a firm one standard deviation higher in alignment retains approximately 398 additional employees, or 6.4% of mean employment. Valued using the mean wage bill per employee, this amounts to approximately RMB 51 million (USD 8 million). Across the interquartile range of alignment, the corresponding difference is about 250 employees, valued at RMB 32 million (USD 5 million).

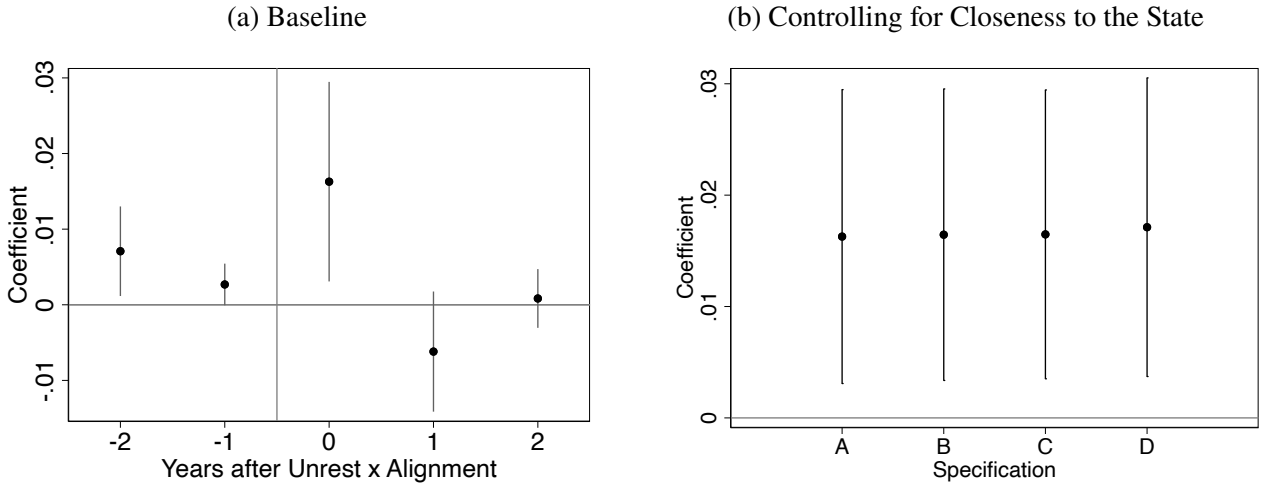
The two main estimates are also similar in order of magnitude. After the Sun Zhengcai scandal, the implied valuation difference between firms at the 75th and 25th percentiles of alignment is approximately RMB 250 million for a firm of average size. The same alignment gap implies an additional annual wage bill of approximately RMB 32 million during a year of elevated unrest. Capitalized at a 10% rate, this flow corresponds to approximately RMB 320 million, comparable to the valuation difference. Retained employees continue to produce output, so the wage bill overstates the firm's net cost. We therefore view this comparison as a descriptive consistency check.

We next examine whether other forms of political embeddedness account for the result. We retain the full lead-and-lag structure in equation (2) and progressively interact every unrest term with state ownership, board political connections, and Party-cell adoption. Figure 4(b) reports the contemporaneous alignment–unrest coefficient across these specifications.[34] Observable political connectedness and state influence therefore do not account for the contemporaneous association. Adding these controls renders the one-year pre-period coefficient statistically insignificant, leaving a significant lead effect only at the two-year horizon.

We also examine whether the employment response reflects differential exposure to economic shocks correlated with unrest. We repeat the lead-and-lag analysis using revenue, profitability, labor productivity, and government subsidies as outcomes, and report the estimates in Figures A.12 and A.13 and Table A.11. We find no systematic differential responses across these outcomes. Two of the 20 interaction coefficients in Table A.11 are significant at the 10% level, approximately the number

[34] We report the corresponding estimates in Table A.6.

Figure 4: Employment Response to Unrest by Public Alignment



Notes: The unit of observation is a firm-year. Figure 4(a) presents a distributed lead-and-lag specification, plotting coefficients on the interaction between city-level labor unrest and firm-level public alignment, measured as the first principal component of Party-related phrase indicators, by event time. Negative event times denote years before unrest, and positive event times denote years after unrest. The specification controls for uninteracted unrest at all leads and lags, alignment, firm fixed effects, and province-year fixed effects. Figure 4(b) reports the contemporaneous alignment–unrest coefficient across four specifications: (A) the baseline; (B) additionally interacting every unrest lead and lag with state ownership; (C) additionally interacting every unrest lead and lag with political connections; and (D) additionally interacting every unrest lead and lag with Party-cell adoption. Standard errors are clustered at the city level. Bars represent 95% confidence intervals.

expected by chance. The employment response also remains positive after controlling for firm-level political risk in the same way as public alignment, as reported in Figure A.15 and Table A.8. These results make it more likely that the employment response reflects costly political support than changes in aligned firms' economic performance.

Taken together, the differential employment response rules in a screening role for public alignment within our framework. Its direction is also consistent with optimal screening: more-aligned firms provide greater support when political conditions deteriorate.

6.3 Additional Empirical Tests

Having used the valuation and employment responses to rule in a screening role within our framework, we now examine whether additional features of firm behavior and treatment are consistent with optimal screening. We consider three implications concerning firms' costs of support, their provision of other

costly activities, and the favorable treatment they receive from the state. Appendix D provides measurement details for each exercise.

Regulatory enforcement actions. Under optimal screening, firms with lower opportunity costs of support are more likely to align. Regulatory scrutiny can reduce the value of business as usual by increasing regulatory risk and disrupting normal operations. In our framework, this lowers c, the opportunity cost of diverting resources toward political support, and makes the aligned contract relatively more attractive. This logic predicts that firms should increase public alignment following regulatory scrutiny.

We test this prediction using an event study around firm-level regulatory enforcement actions:

$$\text{Alignment}_{i,t} = \sum_{k=-2}^{2} \beta_k \, \text{Investigation}_{i,t-k} + \alpha_i + \tau_t + \varepsilon_{i,t}.$$

The outcome is the alignment of firm i in year t, and $\text{Investigation}_{i,t-k}$ denotes lead-and-lag counts of the pooled regulatory-action measure, tracing alignment around regulatory scrutiny. The specification includes firm and year fixed effects, α_i and τ_t, and standard errors are clustered at the firm level. As Figure A.8(a) shows, public alignment increases following regulatory enforcement actions, consistent with regulatory scrutiny lowering firms' opportunity costs of support and making alignment more attractive. Allowing the alignment response to regulatory scrutiny to differ by state ownership, board political connections, and Party-cell adoption attenuates the average response, which remains positive but is estimated less precisely, as Figure A.8(b) shows.

Corporate social responsibility. Optimal screening also implies that aligned firms have lower costs of undertaking actions that advance regime priorities. Many corporate social responsibility (CSR) activities, including employment stabilization, poverty alleviation, environmental compliance, and community programs, may advance these priorities while imposing costs on firms. Alignment should therefore be associated with greater CSR activity if it identifies firms more willing to bear such costs.

We test this prediction using firms' overall CSR scores. Figure A.9 shows that public alignment is

positively associated with CSR, and the relationship remains positive after accounting for year fixed effects, ownership, board political connections, and Party-cell adoption. This pattern is consistent with aligned firms undertaking other costly activities that advance regime objectives.

Government subsidies and procurement. Finally, we examine whether more-aligned firms receive more favorable treatment from the state. Under optimal screening, the aligned contract carries a higher support burden and, when both contracts are chosen, a higher transfer. Government subsidies and procurement contracts provide observable forms of such favorable treatment. The model's transfer prediction compares firms that select different alignment contracts. We therefore estimate this relationship across firms rather than identifying it from firms that switch alignment over time. Firm fixed effects would isolate a switching margin that the static model does not characterize. We interpret these estimates as descriptive associations, not causal effects.

We regress total government subsidies and the number of government procurement contracts received on lagged public alignment. As Figures A.10 and A.11 show, higher lagged alignment is associated with more government subsidies and more government procurement contracts; we report the corresponding estimates in Tables A.9 and A.10. Both associations remain positive and statistically significant after accounting for year fixed effects, ownership, board political connections, and Party-cell adoption. These patterns are consistent with the favorable-treatment margin of optimal screening.

Together, these additional findings are consistent with optimal screening in terms of both allocations and transfers: higher public alignment is associated with costly activities that advance regime objectives and with more favorable treatment.

7 Alternative Interpretations

Our evidence rules in a screening role for public alignment within our framework. This interpretation is subject to two qualifications. First, public alignment need not be the only channel through which the regime screens firms. Private communication, political connections, ownership ties, Party organi-

zations, and other forms of embeddedness may play related roles. Second, annual reports are public and may affect investors, employees, consumers, and other stakeholders.[35] If these effects cause firms with different mean support costs to select different alignment levels, alignment has a screening role under our definition even if the regime does not deliberately induce the sorting. If they instead affect firm values or behavior directly without such sorting, they operate through channels outside the basic model. We therefore consider whether these or other channels can account for our findings without a screening role.

On the first qualification, the public nature of alignment has properties relevant for screening, which may make it useful even beyond purely private forms of firm–regime communication. By using regime language in annual reports, a firm creates a durable record of its political position that is observable to the regime and outside stakeholders. Appendix A.2 provides qualitative evidence that Chinese government agencies systematically read annual reports and, separately, publicize and reward visible political alignment among listed firms, while our empirical results show that investors price the alignment that these reports contain. Unlike private exchanges, public alignment survives personnel turnover, creates a durable record of past communications, and helps outside stakeholders interpret costly firm actions. Public alignment may therefore help sustain that relationship even when some of its terms are negotiated privately, complementing private communication and political connections.

On the second qualification, our interpretation rests on an identifying assumption: outside-audience channels that do not sort firms by mean support cost do not independently generate the joint empirical pattern. To account for the results without a screening role, investor, employee, or consumer responses would have to generate both differential valuation changes when elite political stress is realized and differential employment responses during local unrest without changing the mean support costs of the two alignment groups. We now consider specific alternative explanations.

[35] A growing literature shows that corporate political speech can shape consumer demand, employee behavior, and investor beliefs (Hassan et al., 2019; Bhagwat et al., 2020; Barari, 2024; Conway and Boxell, 2026; Ottonello et al., 2024; Cassidy and Kempf, 2025).

Investor-facing political risk. A first alternative is that public alignment is directed primarily at investors. Investors may interpret regime-specific language as a signal of political risk, future state support, or exposure to political campaigns. This interpretation is related to work showing that firms' political language conveys information to financial markets (Hassan et al., 2019; Ottonello et al., 2024; Cassidy and Kempf, 2025). It could also explain the stock-return result: if investors regard alignment as a sign of dependence on the regime, an event that weakens the regime may reduce the expected value of more-aligned firms.

However, an investor-facing account does not by itself explain why more-aligned firms retain more workers during local unrest. That result concerns a costly real action, not just investors' beliefs about political exposure. If investors price alignment because it identifies firms expected to provide such support, their response instead reinforces the screening interpretation. Investor observability may also explain why screening takes a public form: it allows firms' subsequent support activities to be understood as part of an ongoing relationship with the regime.

Consumers, employees, and other audiences. Public alignment may also target consumers, employees, suppliers, or other non-government audiences. Firms may use regime language to appeal to politically aligned consumers, reassure employees, cultivate legitimacy, or conform to prevailing social expectations. This possibility is related to research showing that firms' political positions can affect consumer demand, labor supply, and stakeholder relationships (Bhagwat et al., 2020; Barari, 2024; Conway and Boxell, 2026; Cassidy and Kempf, 2025; Bombardini and Trebbi, 2025). In China, nationalism and Party-oriented conformity may similarly shape firms' communication with domestic stakeholders (Yue et al., 2024; Lin and Milhaupt, 2021).

These channels may help explain why firms use regime language in public documents, but they do not explain why the same language predicts both lower valuations after an elite political shock and greater employment during local unrest. Such an explanation would need to account for why stakeholder responses become especially important around these political events and why they lead

more-aligned firms to bear greater employment costs. We cannot exclude these channels, yet none directly predicts the joint patterns we observe.

Coordination among firms. Another possibility is that public alignment facilitates coordination among firms. Firms may be more willing to provide costly support if their competitors are expected to do the same, with public alignment serving as a coordination device. This explanation predicts that alignment should vary substantially across competitive markets.

The variance decomposition in Table A.3 does not suggest that simple coordination through common local-market movements is the primary source of alignment variation. With year fixed effects alone, the adjusted R^2 is 18.4%; adding industry, city, and industry-by-city fixed effects raises it by 11.0 percentage points, to 29.4%, so some alignment tracks fixed local and sectoral conditions. Coordination among local competitors, however, would produce common movement within a given local market and year, which industry-by-city-by-year fixed effects would absorb. Adding these fixed effects does not increase the adjusted R^2; it declines slightly, to 28.7%, indicating that they consume degrees of freedom without meaningfully increasing explained variation. Firm fixed effects, by contrast, raise the adjusted R^2 by a further 21.9 percentage points, to 50.6%. Conditions within local markets, including coordination, therefore account for a relatively modest share of the variation in alignment compared with differences across firms. We note that this exercise does not rule out coordination operating through firm-specific relationships or markets defined more narrowly than industry by city.

Generic political risk or political attention. Yet another possibility is that our measure captures generic political risk, political attention, or policy salience rather than alignment with the regime. This concern arises because annual reports frequently discuss regulation, macroeconomic policy, and government programs. We address it in three ways. First, the measure is constructed from phrases that distinguish CCP publications from standard Mandarin, rather than from generic political or economic terms. Second, the LLM identifies explicit pledges of loyalty or support in 22.4% of reports under the

full-text approach and 24.2% under the keyword-based approach, indicating that in a substantial share of reports the language conveys approval rather than neutral description (see Appendix C.1). Third, the measure is only moderately correlated with existing measures of firm-level political risk (Hassan et al., 2019), and our main empirical results are robust to controlling for a firm-level political-risk measure constructed in the same manner (see Section C.2). These results suggest that the measure captures a distinct dimension of firm speech: the use of regime-specific language in public disclosures.

Government business. A distinct possibility is that firms that transact with the state adopt its language as part of that commercial relationship, so that alignment reflects government business rather than screening. Such a relationship could produce public alignment in two ways. The first is mechanical: government terminology enters the report as technical or descriptive language accompanying the business itself. The second is expressive: firms that depend on government business may adopt approving language toward the regime.

Our finding that alignment conveys approval rather than neutral description (see Section C.1) cuts against the mechanical channel. The expressive channel corresponds to government consumption in our framework. If firms adopt approving language because the regime values it and rewards it with business or other benefits, the firm receives favorable treatment while the regime obtains public approval. This channel may coexist with screening, but it does not by itself create a screening role.

Within the basic model, Proposition 1 implies that government consumption alone cannot generate our two central differential findings: absent a screening role, alignment predicts neither differential mean payoff responses nor differential support responses when a bad state is realized. The employment response is not accompanied by systematic differential changes in revenue or subsidies, the outcomes through which additional government business would most directly appear, so the pattern does not reflect aligned firms simply winning more state business when unrest occurs. Both results also survive conditioning on the measures of state ties we examine next, which proxy for the depth of a firm's commercial and institutional relationship with the state.

Observable closeness to the state. A related concern is that public alignment proxies for other forms of closeness to the state. State-owned firms, politically connected firms, and firms with Party cells may use more regime language and also respond differently to political shocks. This possibility is consistent with extensive evidence that political connections and state ties affect firm value, access to resources, and corporate behavior (Fisman, 2001; Lin and Milhaupt, 2021; Chen et al., 2023; Xie et al., 2022; Liu and Zhang, 2019; Zhang, 2019; Fan, 2021).

We address this concern directly in the empirical analysis. In our stock-return exercise, we allow the response to the Sun Zhengcai scandal to vary with state ownership, board political connections, and Party-cell adoption. In the unrest exercise, we similarly allow the employment response to vary along these dimensions. The main results remain: more-aligned firms experience larger valuation losses after the scandal and retain more workers during local unrest.

These other ties may themselves convey information and serve related screening roles. We do not claim that public alignment is the regime's only screening channel. Rather, public alignment is a distinct and observable margin that contains information beyond formal ownership, board political connections, and Party organizational presence.

Underlying firm and managerial characteristics. A final concern is that public alignment reflects underlying firm or managerial characteristics. Larger firms, firms in strategic sectors, financially constrained firms, or firms with different growth prospects may be more likely to use Party language and may also respond differently to political events. Public alignment may also reflect managers' political preferences.

Some of these characteristics are not alternatives to a screening role; they may instead be sources of it. Our framework allows firms' productivity, financial constraints, and outside opportunities to determine their costs of providing support. If these characteristics cause firms with different mean support costs to select different alignment levels, alignment has a screening role under our definition. Under optimal screening, lower-cost firms select higher alignment. The concern that remains is

narrower: whether firm characteristics or managerial preferences generate our two main results for reasons unrelated to the firm–regime relationship.

Several features of the design speak to this concern. First, the stock-return exercise identifies its effect from high-frequency movements around a sudden, unanticipated political event, with alignment measured from the last annual report released before the event; because alignment is predetermined and the window is narrow, the result is difficult to attribute to slow-moving firm characteristics or to reverse causality. Second, the unrest analysis includes firm and province-by-year fixed effects, which absorb all time-invariant firm characteristics (including managers' fixed political preferences) as well as any province-level shock common to firms in a given year. Third, we find no differential responses in revenue, profitability, labor productivity, or subsidies, the outcomes that a characteristics-based account (in which aligned firms are simply larger, more constrained, more favored, or on different trajectories) would be expected to move. In addition, both main results are robust to controls for observable political embeddedness and the political-risk measure discussed above.

An interpretation based on firm or managerial characteristics alone would therefore have to survive all of these checks and still account for the two responses we document. Fixed characteristics, managerial preferences, and reverse causality do not readily deliver this joint pattern.

None of these interpretations needs to be ruled out for public alignment to serve a screening role. Public alignment may also have consumption value for the regime, affect outside audiences, and operate alongside private communication or other forms of political embeddedness. We aim only to rule in a screening role: alignment separates firms with different mean costs of providing support. This conclusion does not establish that the regime deliberately induced the sorting or that screening is alignment's only function.

8 Conclusion

Why do firms publicly align themselves with autocratic regimes through corporate speech? We show that such speech can serve a screening role. When the regime values costly firm support but cannot

observe firms' costs of providing it, linking alignment to support burdens and favorable treatment can induce firms to self-select.

We develop a unified contracting framework in which public alignment can arise as noise, government consumption, or screening. Absent screening, alignment cannot predict differences in firms' mean payoff responses or costly actions when political stress materializes. Under optimal screening, lower-cost firms align, bear greater support burdens, and receive higher transfers. We evaluate these predictions using a transparent, text-based measure of public alignment constructed from the annual reports of listed Chinese firms.

Two results rule in a screening role for public alignment, while allowing that alignment and support may also have consumption value for the regime: more-aligned firms experience larger valuation losses after an adverse political event, and they increase employment more, or reduce it less, during local labor unrest—a costly form of support. Further patterns are consistent with optimal screening: firms increase alignment after regulatory investigations, and more-aligned firms have higher corporate social responsibility scores and receive more government subsidies and procurement contracts. Our two main results remain after accounting for state ownership, political connections, Party cells, political risk, and other observable forms of political embeddedness.

Our interpretation has limits. We observe public speech and firm behavior, but not the full set of private agreements, discretionary benefits, or continuation values that may sustain the firm–regime relationship. Public alignment may also affect audiences other than the regime, and it need not be the only channel through which the regime screens firms. The evidence therefore rules in a screening role for public alignment, but does not establish that screening is its only function.

Several questions remain open. One is when public speech becomes informative about future behavior, which depends on firms' support costs, the benefits attached to alignment, and the extent of heterogeneity across firms. A second is how public alignment interacts with other forms of political engagement, including lobbying, investment, ownership, and internal governance. A third is whether similar mechanisms operate outside autocracies. Although formal commitment is generally stronger

in democracies, governments may still value firm support during periods of political stress, and public speech may help identify firms willing or able to provide it.

More generally, where formal commitments are weak, firm speech may do more than communicate preferences or beliefs. By making firms' political positions observable and durable, public alignment can help regimes identify firms, assign support burdens and favorable treatment, and sustain cooperation over time.

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Appendix

A Additional Background

A.1 Institutional Process Behind Annual Reports

This appendix describes how annual reports are produced inside listed Chinese firms. This background helps establish that annual-report language is a form of public firm communication, rather than unstructured or incidental text.

Chinese disclosure rules assign responsibility for periodic reports, including annual reports, to the firm's top decision-makers. Directors and senior managers must provide formal confirmations regarding the truthfulness, accuracy, and completeness of the annual report, which places ultimate responsibility at the top of the organization (China Securities Regulatory Commission (CSRC), 2017). The board secretary also plays a central role in organizing and coordinating information disclosure, including periodic reporting (China Securities Regulatory Commission (CSRC), 2021).

One concrete example is the annual report publication process at Zhongzhu Medical Holding Co., Ltd., whose internal disclosure rules codify a cross-departmental process for preparing periodic reports (Zhongzhu Medical Holding Co., Ltd., 2025). Business units and controlled subsidiaries provide the underlying materials. The board secretary's office coordinates collection and drafting, circulates the draft to directors for review, and organizes formal disclosure after board deliberation. The finance function prepares the financial statements and related materials and submits them to the board secretary or board office.

This institutional process also makes clear that the political language we study is not mechanically required by disclosure rules. The CSRC standard on the content and format of annual reports does not require firms to disclose Party organization or political activity in the corporate-governance section, nor does it require firms to use Party terminology.[36] Firms therefore retain discretion over whether to

[36] China Securities Regulatory Commission, Standard No. 2 on the Content and Format of Annual Reports (2017 revision).

include regime-aligned language in annual reports.

A.2 Regime Observation of Annual Reports

There is also qualitative evidence that the Chinese state reads and responds to firm annual reports. The most direct evidence is that the China Securities Regulatory Commission (CSRC), the main securities regulator, states in its accounting regulatory reports that it assigned dedicated staff to conduct sample reviews of firm annual reports.[37] The CSRC's enforcement overview also lists reviews of listed companies' yearly filings among the regular channels through which it identifies disclosure problems.[38] These materials show that the government reads these filings as part of routine oversight.

These materials document that the state responds to accounting and disclosure issues identified in annual reports through supervision and inquiry, and separately recognizes public communication and visible political alignment. In the same accounting regulatory reports, the CSRC states that it will follow up on accounting and disclosure problems identified through these reviews.[39] At the exchange level, both the Shanghai Stock Exchange and the Shenzhen Stock Exchange maintain public systems for inquiry letters and company replies, including letters tied to annual reports, which document a process in which firms are questioned about their disclosures and required to respond formally.[40] There is also qualitative evidence of organized recognition and rewards for public communication and visible alignment. The China Association for Public Companies, an industry association for listed firms supervised by the securities regulator, publishes annual lists recognizing "best" and "excellent" practices in listed firms' annual-report earnings briefings, evaluating firms on preparation, content quality, investor communication, and dissemination.[41] The same association also publishes annual collections of model Party-building cases among private listed firms and describes these materials as a way to publicize and promote exemplary practices.[42] Taken together, these materials show that

[37] See CSRC, 2023, CSRC, 2024, and CSRC, 2025.
[38] See CSRC enforcement overview.
[39] See CSRC, 2023, CSRC, 2024, and CSRC, 2025.
[40] See Shanghai Stock Exchange, Regulatory Inquiries and Shenzhen Stock Exchange, Inquiry Letters.
[41] See China Association for Public Companies, 2025 "Best and Excellent Practices" for annual-report earnings briefings.
[42] See China Association for Public Companies, *Excellent Party-Building Cases of Privately-Owned Listed Companies*.

the state systematically reads firm disclosures and separately publicizes and rewards visible political alignment among listed firms.

B Theoretical Appendix

B.1 Proof of Proposition 1

Suppose that alignment has no screening role and that the support burdens under the two contracts differ. Without loss of generality, let $s_1 > s_0$. A type-c firm chooses the aligned contract (s_1, t_1) over the nonaligned contract (s_0, t_0) if and only if

$$t_1 - cs_1 \geq t_0 - cs_0 \iff c \leq \frac{t_1 - t_0}{s_1 - s_0} =: \hat{c}.$$

Because both contracts are chosen by a positive measure of types, $\hat{c} \in (\underline{c}, \overline{c})$. We can therefore write

$$s(c) = \begin{cases} s_0, & \text{if } c > \hat{c}, \\ s_1, & \text{if } c \leq \hat{c}. \end{cases}$$

Consequently,

$$\mathbf{Cov}[c, -s(c)] = (s_1 - s_0)\, F(\hat{c})\, [1 - F(\hat{c})]\, [\mathbf{E}[c \mid c > \hat{c}] - \mathbf{E}[c \mid c \leq \hat{c}]] > 0.$$

This contradicts the hypothesis that alignment has no screening role. Therefore, $s_1 = s_0$.

It remains to compare the induced payoffs. When $s_1 = s_0$, the difference between the firm's payoffs under the two contracts is

$$(t_1 - cs_1) - (t_0 - cs_0) = t_1 - t_0,$$

which does not depend on c. If $t_1 \neq t_0$, all types strictly prefer the contract with the higher transfer, contradicting the assumption that both contracts are chosen by a positive measure of types. Thus, $t_1 = t_0$. Hence, for every type, the two contracts assign the same *ex ante* expected payoff and support

burden. Let s and t denote the common burden and transfer. Since alignment has no screening role,

$$\mathbf{E}[t - cs \mid a = 1] = t - s \cdot \mathbf{E}[c \mid a = 1] = t - s \cdot \mathbf{E}[c \mid a = 0] = \mathbf{E}[t - cs \mid a = 0].$$

The two alignment groups therefore have the same mean *ex ante* expected payoff and support burden.

B.2 Proof of Proposition 2

For a type-c firm, the total surplus generated by alignment choice a and support burden s is

$$U_a + V_a s - cs = U_a + (V_a - c)\, s.$$

For each $a \in \{0, 1\}$, this expression is affine in s. Hence, its maximum over $s \in [0, q]$ is attained at either $s = 0$ or $s = q$.

Consider first the aligned contract. Under Assumption 1,

$$V_1 - c \geq V_1 - \overline{c} \geq 0, \qquad \text{for all } c \in [\underline{c}, \overline{c}].$$

Conditional on alignment, the surplus-maximizing support burden is therefore q. Alignment with maximal support generates $U_1 + q\,(V_1 - c)$, which weakly exceeds the surplus U_1 generated by alignment without support.

Now consider nonalignment. Because surplus is affine in s, the optimal nonaligned support burden is either 0 or q. At the maximal support burden, the first inequality in Assumption 1 implies

$$U_1 + q\,(V_1 - c) \geq U_0 + q\,(V_0 - c)\,, \qquad \text{for all } c \in [\underline{c}, \overline{c}].$$

Thus, alignment with maximal support weakly dominates nonalignment with maximal support.

It follows that, for surplus maximization, it is without loss to consider only the two allocation alternatives: nonalignment with no support and alignment with maximal support. The former generates surplus U_0, while the latter generates surplus $U_1 + (V_1 - c)\, q$. Thus, the latter is optimal if and only if

$$U_1 + (V_1 - c)\, q \geq U_0 \iff c \leq V_1 + \frac{U_1 - U_0}{q} =: c^*.$$

This allocation is implementable by a menu satisfying $s_0^* = 0$ and $s_1^* = q$. When both contracts are chosen by a positive measure of types, the cutoff type must be indifferent:

$$t_0^* = t_1^* - c^* q \iff t_1^* - t_0^* = U_1 - U_0 + qV_1.$$

Because F is atomless, the alignment choice of the cutoff type does not affect any *ex ante* allocation or surplus calculation.

B.3 Proof of Proposition 3

By Proposition 2, the maximal surplus generated for a type-c firm is

$$\Phi(c) = \max\{U_0, U_1 + q(V_1 - c)\} = U_0 + q(c^* - c)_+ .$$

The optimal two-contract menu therefore generates expected surplus $\mathbf{E}[\Phi(c)]$. If the regime is restricted to one contract, it must choose the same alignment and support burden for every type. Applying the argument in the proof of Proposition 2 to the average type $\mathbf{E}[c]$, the maximal pooling surplus is $\Phi(\mathbf{E}[c])$. Hence,

$$\Sigma = \mathbf{E}[\Phi(c)] - \Phi(\mathbf{E}[c]) = q\left[\mathbf{E}[(c^* - c)_+] - (c^* - \mathbf{E}[c])_+\right]$$

(i) On the one hand, if $c^* \le \underline{c}$, $\Phi(c) = U_0$ for every $c \in [\underline{c}, \overline{c}]$, implying that

$$\mathbf{E}[\Phi(c)] = U_0 = \Phi(\mathbf{E}[c]) \implies \Sigma = 0.$$

Similarly, if $c^* \ge \overline{c}$, $\Phi(c) = U_1 + (V_1 - c)\, q$ for every $c \in [\underline{c}, \overline{c}]$, implying that

$$\mathbf{E}[\Phi(c)] = U_1 + q(V_1 - \mathbf{E}[c]) = \Phi(\mathbf{E}[c]) \implies \Sigma = 0.$$

On the other hand, if $c^* \in (\underline{c}, \overline{c})$, then Φ is convex and has a strict kink at c^*. Since F has a positive density, it assigns positive probability to types on both sides of c^*. Consequently, Φ is not affine on the convex hull of the support of F, and so equality in Jensen's inequality cannot hold: $\Sigma = \mathbf{E}[\Phi(c)] - \Phi(\mathbf{E}[c]) > 0$, as claimed.

(ii) When $c^* < \mathbf{E}[c]$, we can write

$$\Sigma = q \int_{-\infty}^{c^*} (c^* - c)\, \mathrm{d}F(c) \implies \frac{\partial \Sigma}{\partial c^*} = qF(c^*) \geq 0.$$

When $c^* > \mathbf{E}[c]$, we can write

$$\Sigma = q \int_{c^*}^{\infty} (c - c^*)\, \mathrm{d}F(c) \implies \frac{\partial \Sigma}{\partial c^*} = -q\,[1 - F(c^*)] \leq 0.$$

(iii) Since Φ is convex, a mean-preserving spread of F weakly increases $\mathbf{E}[\Phi(c)]$. However, a mean-preserving spread of F does not change $\Phi(\mathbf{E}[c])$. Thus, $\Sigma = \mathbf{E}[\Phi(c)] - \Phi(\mathbf{E}[c])$ is nondecreasing in mean-preserving spreads of F.

B.4 Proof of Proposition 4

Because F is atomless, the cutoff type's choice does not affect the *ex ante* probability of alignment. Under the surplus-maximizing menu characterized in Proposition 2, the probability of alignment is therefore $F(c^*)$, which is nondecreasing in c^* and strictly increasing while $c^* \in (\underline{c}, \overline{c})$. Moreover, for distributions $F_H \leq F_L$ supported on $[\underline{c}, \overline{c}]$ (i.e., $F_H \succeq_{\text{FOSD}} F_L$), we obtain

$$F_H(c^*) \leq F_L(c^*).$$

Thus, the *ex ante* probability of alignment is nonincreasing under first-order stochastic dominance shifts of F, as claimed.

B.5 Repeated-Game Foundation

We provide a repeated-game foundation for the commitment and surplus-maximization assumptions in the basic model.

Environment. A firm and a regime interact in periods $t = 1, 2, \ldots$ and share a discount factor $\delta \in (0, 1)$. Before period 1 and before the first cost realization, the parties may prescribe a one-time voluntary public transfer $z \in \mathbb{R}$, where $z > 0$ denotes a transfer from the regime to the firm. The transfer is observed before the recurrent interaction begins.

At the beginning of each period, the firm privately observes a cost type $c_t \in [\underline{c}, \overline{c}]$, drawn independently from F. Before alignment is chosen, either party may terminate the relationship; the regime does not observe c_t. If the relationship continues, the firm chooses a publicly observable alignment level $a_t \in \{0, 1\}$. The political state $\theta_t \in \{B, G\}$ is then publicly realized, independently across periods and independently of the current cost type, alignment choice, and public history, with $\Pr[\theta_t = B] = q \in (0, 1)$.

In the bad state, a public randomization device determines whether support is prescribed, and the firm decides whether to provide it. Let $X_t \in \{0, 1\}$ indicate whether support is provided; in the good state, $X_t = 0$. At the end of the period, the parties may make a voluntary public transfer. Let $\tau_t \in \mathbb{R}$ denote the net transfer from the regime to the firm. A positive transfer is paid by the regime and a negative transfer by the firm. A refusal to provide prescribed support or to make a prescribed transfer is publicly observed. Either party may also terminate after any public history. Termination is absorbing and gives both parties a payoff of zero.

The firm's period payoff is

$$\tau_t - c_t X_t,$$

while the regime's period payoff is

$$u_{a_t}(\theta_t) + V_{a_t} X_t \mathbf{1}\{\theta_t = B\} - \tau_t.$$

Neither support nor transfers are externally enforceable.

Our solution concept is stationary perfect public equilibrium (PPE). The public history consists of past alignment choices, political states, public randomizations, support choices, transfers, and termination decisions. A public strategy for the firm is a publicly specified type-contingent rule: after every public history, the firm's actions may depend on its current private type c_t, while the regime's strategy depends only on public history. After an alignment choice that occurs with positive probability, the regime updates its belief about c_t using Bayes' rule; beliefs following a zero-probability alignment may be arbitrary. Strategies are sequentially optimal after every public history and, for the

firm, for every current type.

We focus on stationary PPEs in which, beginning in period 1, support and transfer rules depend on the firm's type only through its alignment choice and the relationship continues on the equilibrium path for every type. The same alignment-contingent rules are used in every period, and the same continuation equilibrium begins next period following every on-path alignment choice and prescribed action. Termination is used only following a publicly observed deviation; off-path continuation play may depend on the public history.

Let W_F and W_R denote the parties' expected payoffs at the beginning of a recurrent period, before the firm observes its type. By stationarity, the same continuation payoffs follow every on-path prescribed action.

Static representation. Define the regime's expected direct value from alignment a by

$$U_a \coloneqq q u_a(B) + (1-q) u_a(G).$$

Conditional on alignment a, let $x_a \in [0,1]$ denote the probability that support is provided in the bad state and denote the expected support burden by

$$s_a \coloneqq q x_a \in [0, q].$$

Let p_a denote the expected current transfer following alignment a and compliance with the prescribed actions.

A menu in the basic model is a pair of contracts

$$\mathcal{M} = \{(s_a, t_a) : a \in \{0,1\}\},$$

where a type-c firm chooses an alignment that maximizes $t_a - c s_a$.

Proposition 5. *Every stationary PPE induces, up to a set of types of measure zero, an alignment and support allocation implementable by a menu in the basic model. If $a(c)$ denotes the induced alignment*

choice, total surplus satisfies

$$W_F + W_R = \frac{1}{1-\delta} \int_{\underline{c}}^{\overline{c}} \left[U_{a(c)} + \left(V_{a(c)} - c \right) s_{a(c)} \right] \, \mathrm{d}F(c).$$

Proof. Suppose first that both alignments are chosen with positive probability. Define the effective transfer

$$t_a \coloneqq p_a + \delta W_F.$$

A type-c firm that chooses a and follows the prescribed actions receives $t_a - cs_a$. Because choosing either alignment is feasible, equilibrium requires

$$a(c) \in \arg\max_{a \in \{0,1\}} \{ t_a - cs_a \}.$$

Thus, the induced allocation is implementable by a menu in the basic model.

If only alignment a^* is chosen with positive probability, define (s_{a^*}, t_{a^*}) as above. For the unused alignment b, choose any $s_b \in [0, q]$ and choose t_b sufficiently low that

$$t_b < \min_{c \in [\underline{c}, \overline{c}]} \left\{ t_{a^*} - cs_{a^*} + cs_b \right\}.$$

The resulting menu implements the same pooling allocation.

Finally, current transfers cancel when the parties' payoffs are added. Hence, with $W \coloneqq W_F + W_R$,

$$W = \int_{\underline{c}}^{\overline{c}} \left[U_{a(c)} + \left(V_{a(c)} - c \right) s_{a(c)} \right] \, \mathrm{d}F(c) + \delta W.$$

Rearranging yields the desired expression. □

Implementation. Because the regime does not observe the firm's type, a different alignment choice cannot be punished and must instead be deterred by incentive compatibility. All remaining deviations—including refusing prescribed support, failing to make a prescribed transfer, and terminating the relationship—are publicly observed.

Proposition 6. *Consider an incentive-compatible menu in the basic model,* $\mathcal{M} = \{(s_a, \widetilde{t}_a) : a \in$

$\{0,1\}$*, and let* $a(c)$ *denote its induced alignment rule. If its expected one-period surplus*

$$\mathcal{S} \coloneqq \int_{\underline{c}}^{\overline{c}} \left[U_{a(c)} + \left(V_{a(c)} - c\right) s_{a(c)}\right] \mathrm{d}F(c) > 0,$$

then its alignment and support allocation can be implemented in a stationary PPE for all sufficiently high δ*.*

Proof. Fix $\lambda \in (0,1)$ and let

$$\widetilde{\Pi}_F \coloneqq \int_{\underline{c}}^{\overline{c}} \left[\widetilde{t}_{a(c)} - c s_{a(c)}\right] \mathrm{d}F(c).$$

Set the initial transfer equal to zero. Following alignment a, prescribe support in the bad state with probability s_a/q and prescribe the end-of-period transfer

$$p_a \coloneqq \widetilde{t}_a - \widetilde{\Pi}_F + \lambda\mathcal{S}.$$

Following either alignment and compliance with the prescribed actions, the same continuation equilibrium begins next period. Any observed failure to follow the prescribed support or transfer rule leads to no further transfer that period and permanent termination.

The firm's *ex ante* expected current payoff is $\lambda\mathcal{S}$, while the regime's is $(1-\lambda)\,\mathcal{S}$. Their stationary continuation payoffs are therefore

$$W_F = \frac{\lambda\mathcal{S}}{1-\delta} \quad \text{and} \quad W_R = \frac{(1-\lambda)\,\mathcal{S}}{1-\delta}.$$

The firm's effective transfer following alignment a is

$$t_a^\delta = p_a + \delta W_F = \widetilde{t}_a - \widetilde{\Pi}_F + \frac{\lambda\mathcal{S}}{1-\delta}.$$

The effective transfers therefore differ from the original transfers by the same constant for both alignments. The alignment rule $a(c)$ remains incentive compatible. Moreover, a type-c firm's payoff from continuing is

$$\widetilde{t}_{a(c)} - c s_{a(c)} - \widetilde{\Pi}_F + \frac{\lambda\mathcal{S}}{1-\delta}.$$

Because the menu is finite and the type space is compact, $\widetilde{t}_{a(c)} - cs_{a(c)} - \widetilde{\Pi}_F$ is uniformly bounded below over c. This payoff is therefore positive for every type when δ is sufficiently high. Thus, no firm type prefers to terminate before choosing alignment. Since the regime does not observe c before alignment, it receives $W_R > 0$ from continuing and likewise prefers not to terminate.

Let D_F, D_R bound the parties' current-period gains from any public one-shot deviation other than a change in alignment, at any history following either alignment at which the relationship remains active and, for the firm, uniformly over its current type. These bounds can be chosen independently of δ because the type space is compact and the primitive payoffs and prescribed transfers are finite. Because transfers are voluntary, neither party can compel a payment from the other, so the only potentially profitable deviation at the transfer stage is for the prescribed payer to withhold some or all of its payment.

A deviation that triggers termination gives the deviator a continuation payoff of zero. The remaining enforcement conditions therefore hold whenever

$$D_F \leq \frac{\delta\lambda\mathcal{S}}{1-\delta} \quad \text{and} \quad D_R \leq \frac{\delta\,(1-\lambda)\,\mathcal{S}}{1-\delta}.$$

Because $\mathcal{S} > 0$ and $\lambda \in (0,1)$, both inequalities hold for all sufficiently high δ. They rule out deviations in support, transfers, and termination at every active public history following either alignment and uniformly over the firm's current type. The preceding participation argument rules out termination before alignment, while incentive compatibility rules out type-contingent deviations in alignment.

Because termination is unilateral and absorbing, the termination continuation is itself a PPE. The one-shot deviation principle therefore establishes sequential optimality after every public history. Bayes' rule determines beliefs following alignment choices made with positive probability. Beliefs following a zero-probability alignment may be arbitrary and do not affect the construction because the same continuation and enforcement bounds apply following either alignment. □

Pareto efficiency and surplus maximization. Following Levin (2003), the one-time initial transfer separates the allocation of surplus from its distribution. We evaluate Pareto efficiency within the class defined above and at the beginning of the relationship: a stationary PPE is *ex ante Pareto efficient* if no agreement consisting of an initial transfer followed by another stationary PPE in this class makes both parties weakly better off and at least one strictly better off.

Proposition 7. *Every* ex ante *Pareto-efficient stationary PPE maximizes total surplus among stationary PPEs in this class. If a surplus-maximizing menu in the basic model generates positive expected surplus, then, for all sufficiently high δ, every* ex ante *Pareto-efficient stationary PPE induces a surplus-maximizing menu in the basic model.*

Proof. Consider a stationary PPE with one-period surplus $\mathcal{S}$ and *ex ante* payoffs (P_F, P_R), including any initial transfer. By Proposition 5,

$$P_F + P_R = \frac{\mathcal{S}}{1-\delta}.$$

Suppose another stationary PPE has one-period surplus $\widehat{\mathcal{S}} > \mathcal{S}$ and recurrent continuation payoffs $(\widehat{W}_F, \widehat{W}_R)$. Let

$$\Delta := \frac{\widehat{\mathcal{S}} - \mathcal{S}}{1-\delta} > 0$$

and choose $\varepsilon \in (0, \Delta)$. The following payoff pair sums to $\widehat{\mathcal{S}}/(1-\delta)$:

$$P'_F = P_F + \varepsilon \quad \text{and} \quad P'_R = P_R + \Delta - \varepsilon$$

It can therefore be obtained by preceding the second equilibrium with an initial net transfer

$$z = P'_F - \widehat{W}_F.$$

The original payoffs P_F and P_R are nonnegative because either party can otherwise terminate. Hence P'_F and P'_R are positive. The payer therefore prefers to make the prescribed initial transfer and continue rather than refuse and receive zero. The recipient also prefers to continue because the second equilibrium gives it a nonnegative continuation payoff. Any failure to make the transfer exactly

as prescribed is followed by termination, and the transfer does not change continuation play. The resulting agreement strictly improves both parties' payoffs, contradicting *ex ante* Pareto efficiency. Hence every *ex ante* Pareto-efficient stationary PPE maximizes surplus among stationary PPEs in this class.

Finally, Proposition 5 implies that no stationary PPE in this class generates more one-period surplus than a surplus-maximizing menu in the basic model. By Proposition 6, that menu is implementable for sufficiently high δ whenever its surplus is positive. The two maxima therefore coincide. □

The repeated interaction therefore provides a foundation for focusing on surplus-maximizing menus in the basic model. Within the class considered here, stationarity reduces the allocation problem to the choice of alignment and support, continuation values sustain voluntary support and transfers, and the initial transfer divides the resulting surplus. Once Pareto-dominated stationary relationships are ruled out, the repeated interaction selects the surplus-maximizing allocation for sufficiently patient parties.

C Validating the Alignment Measure

This appendix provides two validation exercises for our baseline keyword-based measure of public alignment. First, we benchmark it against an alternative measure constructed using a large language model (LLM). Our baseline measure is intentionally transparent: it counts a fixed set of 50 regime-specific phrases. This transparency comes at the cost of limited contextual interpretation. An LLM-based approach has the opposite tradeoff. It is less transparent and harder to audit, but can use surrounding text to incorporate context, tone, and semantics. We therefore implement an independent LLM measure of public alignment and compare it to our keyword-based measure. The two approaches produce highly correlated firm-level measures, increasing confidence that the baseline measure captures public alignment rather than incidental use of regime-related language. We also use the LLM to characterize the context in which alignment language appears, showing that firms deploy this language in an overwhelmingly positive, non-critical tone and that explicitly political statements

typically take the form of loyalty pledges.

Second, we examine whether our measure is proxying for existing concepts of political risk. To test this possibility, we implement a standard text-based measure of political risk in the same corpus and compare it to our measure of public alignment. The two measures are not highly correlated. This pattern suggests that alignment captures a conceptually distinct dimension of firm political behavior: not firms' exposure to political uncertainty or risk, but their public displays of loyalty to the regime.

C.1 LLM-Based Validation

C.1.1 An LLM-Based Measure of Alignment

We score annual reports with a locally run version of DeepSeek, a large bilingual language model with strong Chinese-language capabilities.[43] We call the model at temperature zero for reproducibility, so each prompt yields the model's modal response. Because we run the model locally, no content filter removes politically sensitive passages. A single prompt asks the model to perform three tasks on each passage of text. It rates the *extent* of political alignment on a 0–100 scale, rates the *sentiment* of any political language on a numerical scale running from critical to supportive, and identifies the *context* in which political language appears (the firm's own business, its sector, the macroeconomy, or the country's political environment). Within the political-environment context, the model further records whether each statement is a *pledge* of loyalty or support, as opposed to a descriptive statement about politics. Bundling the tasks in one call allows the model to classify alignment, sentiment, and context using the same interpretation of the underlying text. The full prompt is reported in the replication package.

We implement the measure in two ways. The *full-text LLM approach* feeds the entire annual report to the model. The *keyword-based LLM approach* restricts the model to sentences that contain at least

[43] We use DeepSeek-V3.2 (`deepseek-ai/DeepSeek-V3.2` on Hugging Face) and run it locally; we downloaded the model on April 7, 2026. We choose it because it is a strong bilingual model that handles Chinese-language text well. In preliminary tests, several general-purpose models optimized for English-language text, including models by OpenAI, Google, and Anthropic, produced inconsistent classifications on these reports, and many outputs were judged to be of low quality in human review.

one of our 50 phrases.[44] The full-text LLM approach is the more demanding test, since it does not use our phrase list and therefore provides an independent benchmark. The keyword-based LLM approach is closer to our measure by construction and isolates how the model reads the specific passages flagged by our phrases.

Before scoring, we strip the financial-statement section of each report, which consists of audited tables and boilerplate with little substantive political content,[45] and split the remaining text into chunks of 25,000 characters with a 250-character overlap so that no sentence is cut across a boundary. We aggregate the chunk-level scores to the report level. Alignment extent and sentiment are averages across chunks, context counts are sums, and the dominant context is the one with the highest count. The pledge fraction is the share of political-environment statements that are pledges.[46]

Concretely, the prompt elicits sentiment as a numerical score from -100 to 100, anchored at -100 for language critical of policy or resistant to regulation, 0 for factual or compliance-related mentions, and 100 for enthusiastic support or expressions of loyalty. A chunk in which the model detects no political language is scored 0, the same value as a politically neutral chunk. The report-level score is the unweighted mean of the chunk scores, so a short final chunk carries the same weight as a full 25,000-character chunk. We classify a report as positive, neutral, or negative according to whether this mean is positive, exactly zero, or negative. Because chunks without political language score zero, a report is neutral when the model detects no political language in any chunk. Offsetting positive and negative chunks would also produce a zero mean, but no report in our data mixes them, which is also why weighting chunks by length leaves the classification unchanged.

[44] We identify keyword-bearing sentences by exact string match. In a sample of 200 reports, exact matching agrees with `jieba`-based detection for about 95% of reports. Where they differ, `jieba` has merged a phrase into a longer token, so exact matching recovers slightly more occurrences.

[45] We strip this section primarily to keep the analysis computationally feasible. It is unlikely to affect the results, since the financial statements rarely contain any of our 50 phrases.

[46] We drop reports with empty source files and the small number of text chunks that exceed the model's context-length limit (26 chunks across 5 reports). These exclusions are rare and do not materially affect the firm-year measures.

C.1.2 Correlation with the Keyword Measure

The LLM and our main keyword measures are strongly correlated. Across the matched sample of about 45,000 firm-years over 2003–2021, the full-text LLM alignment score correlates with our measure at $r = 0.583$. Because this score never sees our phrase list, the correlation provides evidence that the keyword measure meaningfully captures political alignment. Restricting the LLM to keyword-bearing sentences raises the correlation to $r = 0.662$, as expected when both measures are based on the same passages. Because both series trend upward over the sample period, these pooled correlations partly reflect common time variation that the main specifications absorb. Residualizing both measures with respect to firm and year fixed effects isolates the variation used in the main specifications, and the two scores remain correlated at $r = 0.346$ for the full-text version and $r = 0.475$ for the keyword-restricted version. The full-text and keyword-based LLM scores correlate with each other at $r = 0.612$. Figure A.1 plots these relationships.

The alignment measures also display similar time trends. Figure A.2 shows that mean LLM alignment rises across the sample period under both approaches, tracking the increase in the keyword measure documented in Section 5.

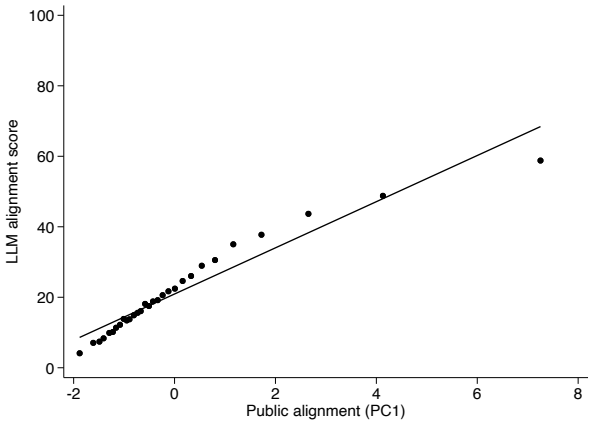
Two points follow. First, the keyword measure appears to capture political alignment: an independent semantic measure recovers similar firm-year variation. Second, the correlation is different from one, indicating that the two approaches capture related but distinct sources of variation. We therefore use the keyword measure as our baseline because it is transparent, reproducible, and fixed across specifications. The LLM measure provides a complementary validation exercise, adding semantic information at the cost of lower interpretability and exact replicability.

C.1.3 Sentiment and Context of Alignment Language

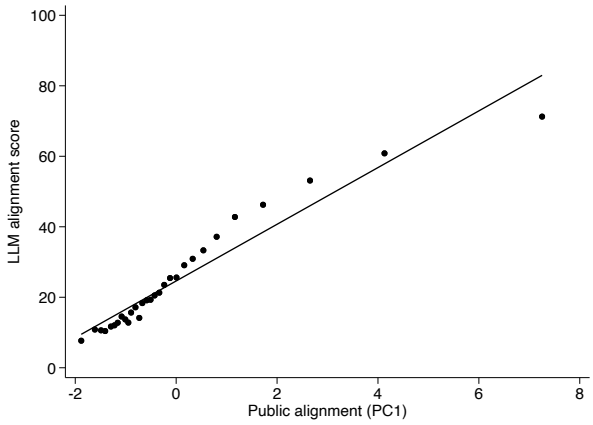
The LLM allows us to characterize how political language is used, not only whether it appears. Two patterns emerge, both of which support interpreting the measure as public alignment rather than

Figure A.1: LLM Alignment Score versus the Keyword Measure

(a) Full-text LLM approach

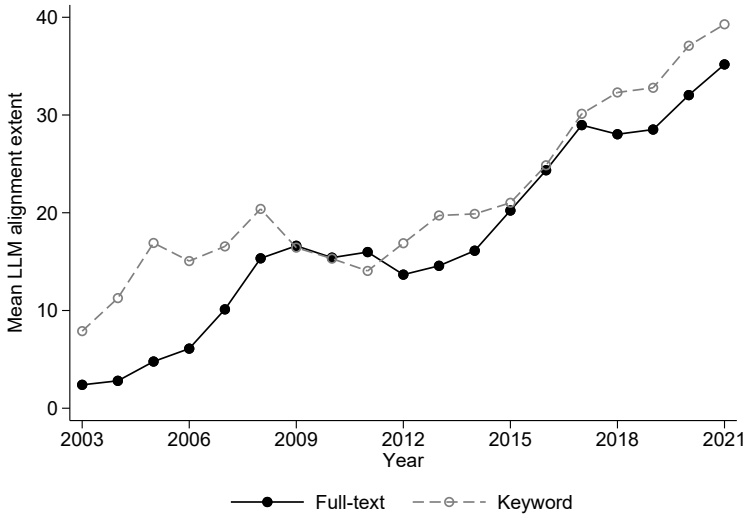


(b) Keyword-based LLM approach



Notes: Binned scatter plots of the LLM alignment score (vertical axis, 0–100) against our keyword-based alignment measure, PC1 (horizontal axis, standardized). Figure A.1(a) uses the full-text LLM score, which does not use our phrase list. Figure A.1(b) restricts the LLM to sentences containing one of the 50 phrases. Pearson correlations are $r = 0.583$ (full-text) and $r = 0.662$ (keyword).

Figure A.2: Mean LLM Alignment over Time



Notes: Mean LLM alignment extent by year, full-text and keyword-based LLM approaches.

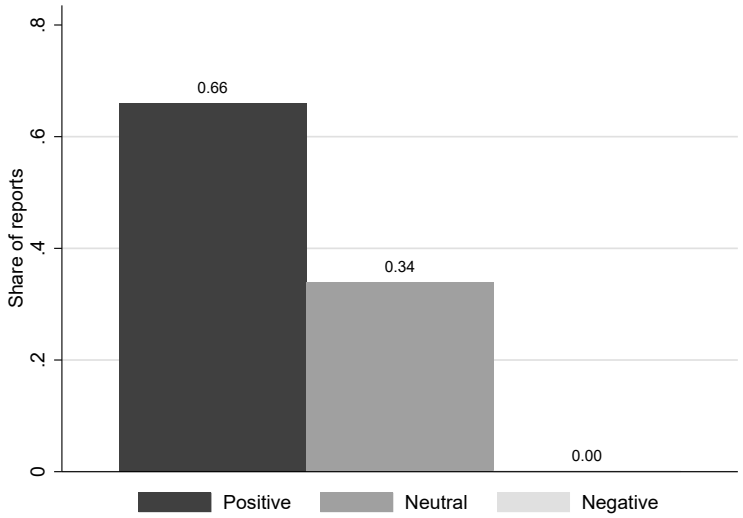
generic references to politics.

First, political language is overwhelmingly positive in tone. The full-text LLM approach classifies 61.4% of reports as having positive political sentiment and 38.6% as neutral. A report is classified as neutral almost exclusively when the LLM finds no political language in it, so the neutral share reflects reports with no detected political content rather than politically neutral reports. Whenever the full-text LLM approach detects political language, it scores it positive almost without exception. The keyword-based LLM approach classifies 66.7% as positive and 33.3% as neutral. Negative sentiment is essentially absent under either approach, below 0.1%. The two approaches agree on sentiment for 79.9% of reports. Sentiment also rises with phrase use. A logit regression of the positive-sentiment indicator on the log of total Party-phrase occurrences yields an odds ratio of 4.48 ($p < 0.001$). The corresponding Spearman rank correlation is $\rho = 0.41$. Figure A.3 shows the sentiment distribution under the keyword-based LLM approach.

Second, public alignment is approving rather than neutral description, and a substantial minority of reports contain explicit pledges of loyalty. Within political-environment passages, the LLM flags pledges of loyalty or support in 22.4% of reports under the full-text LLM approach and 24.2% under the keyword-based LLM approach. The mean pledge fractions are 0.139 and 0.183, respectively.

Both patterns are stronger under the keyword-based LLM approach than under the full-text LLM approach. The keyword filter selects passages with a higher share of positive sentiment and a higher pledge fraction, consistent with the 50 phrases selecting passages with more concentrated and more loyalty-oriented political language. This pattern is consistent with the intended interpretation of the keyword measure. Where firms invoke this language in a political context, they do so in an overwhelmingly positive tone, and their explicitly political statements often take the form of loyalty pledges rather than criticism.

Figure A.3: Sentiment of Political Language (Keyword-Based LLM Approach)



Notes: Share of reports classified as having positive, neutral, or negative political sentiment under the keyword-based LLM approach. Negative sentiment is below 0.1%. Under the full-text LLM approach, the positive share is 61.4% and the neutral share is 38.6%.

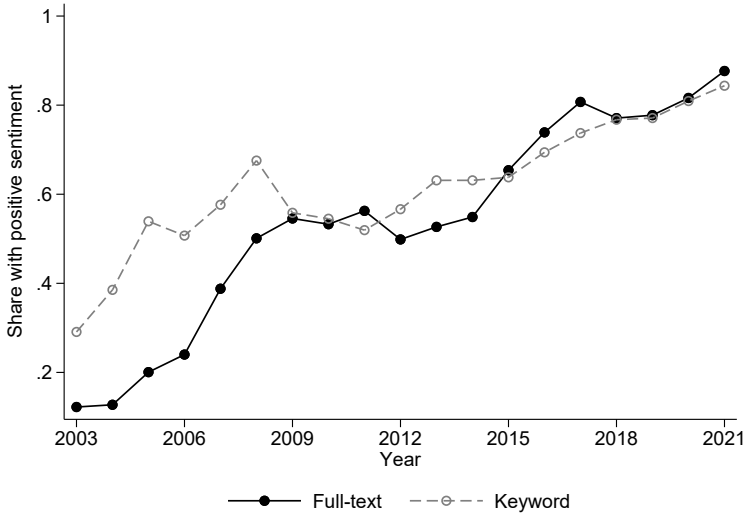
C.1.4 Trends over Time

The sentiment and context of alignment language shifted over the sample period. Under the full-text LLM approach, the share of reports with positive political sentiment rises from 12.2% in 2003 to 87.7% in 2021. Under the keyword-based LLM approach, it rises from 29.1% to 84.4% over the same period (Figure A.4). The dominant context also shifts toward explicit politics: the share of reports whose political language is mainly about the country's political environment, rather than the firm's business, sector, or the macroeconomy, rises over time under both approaches (Figure A.5). Both shifts accelerate after roughly 2012, consistent with a broader increase in ideological emphasis during the Xi Jinping era and with the rise in the keyword measure documented in the main text.

C.2 Distinctiveness from Political Risk

Does our measure capture alignment with the regime or a firm's exposure to political risk? Annual reports discuss regulation, policy, and macroeconomic conditions, so firms facing greater political risk may use more political language for reasons unrelated to alignment. To separate the two, we build

Figure A.4: Positive Political Sentiment over Time

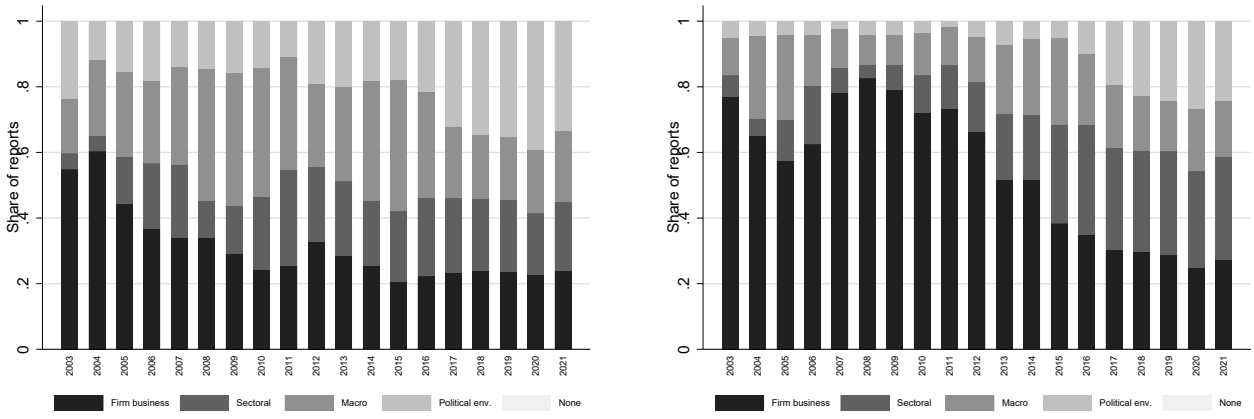


Notes: Share of reports with positive political sentiment by year, full-text and keyword-based LLM approaches.

Figure A.5: Dominant Context of Political Language over Time

(a) Full-text LLM approach

(b) Keyword-based LLM approach



Notes: Shares of reports by dominant context (firm business, sectoral, macroeconomic, political environment) and year under the full-text and keyword-based LLM approaches.

a firm-level measure of political risk for Chinese annual reports following the method of Hassan et al. (2019) and show that it is only moderately correlated with public alignment.

C.2.1 A Political-Risk Measure for Chinese Firms

When a firm is exposed to political risk, its disclosures tend to discuss political topics using language related to risk and uncertainty. Hassan et al. (2019) operationalize this idea as a firm-level measure for U.S. firms. They scan each earnings-call transcript for political vocabulary that appears near words such as "risk" and "uncertainty," and record the share of the transcript made up of such passages. We construct an analogous measure for Chinese annual reports in three steps.

First, we assemble a dictionary of political vocabulary. We take the titles of Chinese Wikipedia entries from 2019, label each as political or non-political with a language model, segment both groups into words with `jieba`, and keep the words that are far more common in political than in non-political text. A word qualifies if it is at least ten times more common in political than in non-political articles. This procedure yields 3,118 terms; the most distinctive include the names of Party organs and offices, such as "Party group" and "municipal Party secretary."

Second, we assemble a short list of risk and uncertainty words. We query a language model ten times for synonyms of the Chinese words for "risk" and "uncertainty" and keep the eleven single-word terms that appear in all ten queries, among them the words for crisis, threat, and turmoil.

Third, we score each annual report. For every word in the report that belongs to the political dictionary and falls within ten words of a risk or uncertainty word, we add the word's distinctiveness weight, then divide the total by the number of words in the report,

$$\text{PRisk}_{it} = \frac{1}{B_{it}} \sum_{b=1}^{B_{it}} \mathbf{1}[b \in \mathcal{P}]\, \mathbf{1}[\min_r |b - r| \leq 10]\ \text{DF-ratio}(b),$$

where b indexes the B_{it} words of the report, $\mathcal{P}$ is the political dictionary, r indexes the positions of risk and uncertainty words, and DF-ratio(b) is how much more common word b is in political than in non-political text. For a report that contains no risk or uncertainty word, we take the minimum

over the empty set to be infinite, so every indicator equals zero and $\text{PRisk}_{it} = 0$. The weight is larger for words that are more exclusively political, so a passage built around a core Party term contributes more than one built around a word that is only mildly political. The replication package reports both dictionaries and the construction code.

Our measure follows the same logic as Hassan et al. (2019) but differs in five respects, each motivated by the Chinese-language setting or by data availability. Table A.1 lists each choice and the reason for it.

C.2.2 Public Alignment Is Not Political Risk

Public alignment and political risk are only moderately correlated. Across about 44,500 firm-years, the Pearson correlation is 0.28 using a ten-word window and 0.24 using a five-word window. A firm can echo Party rhetoric while reporting little political risk, and a firm can report substantial political risk while using little Party language. Public alignment therefore captures a dimension of firm speech that is distinct from political-risk exposure.

C.2.3 Controlling for Political Risk

Public alignment could still shape our main results through its correlation with political risk. To assess this possibility, we re-estimate the stock-return response to the Sun Zhengcai scandal (Section 6.1) and the employment response to local unrest (Section 6.2) controlling for political risk, which we enter in each specification in the same way as public alignment. Figure A.16 and Table A.7 report the scandal results. Political risk is itself a strong negative predictor of post-event returns, as expected if more politically exposed firms lose value when the regime weakens. Public alignment remains a separate negative predictor that political risk does not absorb.

Figure A.15 and Table A.8 report the unrest results. The interaction between alignment and contemporaneous unrest stays positive and statistically significant across specifications, while the interaction between political risk and unrest is small. More-aligned firms increase employment during

Table A.1: Political-Risk Measure: This Paper versus Hassan et al. (2019)

	Hassan et al. (2019)	This paper	Reason for our choice
Source text	Earnings-call transcripts	Annual reports	Chinese listed firms have no comparable earnings calls. Annual reports are the standardized public corpus, and the same text our alignment measure uses.
Token unit	English word pairs (bigrams)	Single Chinese words	English-language text measures often use two-word phrases because individual English words can be too generic to identify a topic (Hassan et al., 2019). In Chinese, by contrast, segmented terms are often multi-character words or phrases that are already relatively specific. We therefore use individual segmented Chinese terms as the unit of analysis.
Political dictionary	Set difference of training libraries	Frequency-ratio threshold over Wikipedia	No Chinese textbook or newspaper training corpus is available. A set difference drops core terms that appear even once in a non-political article, so we keep words far more common in political text instead.
Risk synonyms	Oxford English Dictionary	Language-model consensus	We do not have an authoritative Chinese synonym source directly comparable to the Oxford English Dictionary. We therefore use consensus across ten independent model queries.
Term weight	Frequency in the political corpus	Document-frequency ratio	Hassan et al. (2019) proxy a word's political distinctiveness with its frequency. Our frequency ratio, how much more common a word is in political than non-political text, measures distinctiveness directly.

local unrest even after we control for political-risk exposure.

Public alignment therefore captures variation in firm behavior that political risk does not explain.

D Measurement: Data Details

D.1 CSMAR: Firm Characteristics

We obtain firm-level accounting, ownership, and market variables from the China Stock Market & Accounting Research (CSMAR) database. The database provides annual data from financial statements and balance sheets for listed firms, from which we obtain profit and sales, available for 2004–2021, and the number of employees, available for 2003–2020. Ownership data cover 2003–2020. We classify each firm as state-owned, private, or foreign using these ownership variables. Stock-market data are available at the daily frequency from January 2015 through December 2022. We use these data to compute weekly stock returns and abnormal returns.

D.2 WIND: Annual Reports and Firm Information

We use the WIND Financial Database to construct a firm-year panel for listed firms on the Shanghai Stock Exchange and the Shenzhen Stock Exchange over 2003–2021, covering 3,701 firms as of 2021. From WIND, we download the full set of annual reports and apply the keyword-search procedure described in Section 5 using the phrases listed in Table 1. Beyond annual-report text, WIND supplies additional firm information used for sample construction and controls, including each firm's industry, registered location, listing date, legal name, and English name. WIND also provides the text of firms' corporate-charter amendments, from which we construct the Party-cell indicator. The indicator equals one beginning in the first year in which a firm's charter amendment mentions establishing a Party cell and in every subsequent year.

D.3 WIND: Government Subsidies

We obtain data on government subsidies from the WIND Financial Database, covering listed A-share firms at the firm-year level over 2007–2023. For each firm-year, WIND reports total government subsidies from the government-grants disclosure schedule of the annual report, together with two income-statement line items: other income and government subsidies recorded under non-operating income.

Our subsidy measure is the disclosure-schedule total, following prior studies that measure government subsidies to Chinese listed firms from their financial disclosures (Lee et al., 2014; Branstetter et al., 2023). When it is missing, we instead use the sum of the two income-statement line items. We sum the two lines because the 2017 revision of Chinese Accounting Standard No. 16 (Government Grants) moved subsidies related to firms' daily activities from non-operating income to the newly created other-income line, so that no single line item is comparable across the reform, while the sum of the two approximates the disclosure-schedule total in both regimes. Firm-years not covered by WIND are treated as missing rather than zero.

D.4 CCGP: Government Procurement

We obtain data on government procurement contracts from the China Government Procurement network (CCGP), the Ministry of Finance's official platform for tender and award announcements.[47] Matching listed firms' names against all published announcements yields 410,104 firm-contract documents for 2013–2020; we exclude failed or annulled tenders.

The matching uses each firm's full and abbreviated names. We classify each firm-contract mention by the firm's role, using the announcements' structured fields and the text surrounding the mention: winning supplier, procuring buyer, losing candidate, procurement agent, manufacturer of a supplied product, or incidental mention (for example, a firm named in a bank-account field or a street address). Name matches inconsistent with the firm's registered full name are discarded. In an audit of a random

[47] http://www.ccgp.gov.cn

sample of contracts, approximately 97% of mentions classified as winning suppliers are correct. Our procurement measure is the number of contracts a firm wins as an identified supplier in a given year, set to zero for listed firms that win no contracts. We rely on contract counts rather than award amounts because award amounts are missing for a substantial share of announcements and cannot be extracted reliably from the rest.

D.5 CLB and China Strikes: Labor Unrest

The main unrest variable in Subsection 6.2 is the number of labor-unrest events in the firm's home city and year. We construct this variable from two event-level datasets on worker collective action in China: *China Strikes* (Elfstrom, 2017), which covers 2003–2011, and China Labour Bulletin (CLB) (China Labour Bulletin, 2019), which covers 2012–2022. Both datasets compile reports from workers, media accounts, and other publicly available sources. CLB was developed using China Strikes as a template, and the two sources provide consistent coverage across the 2011–2012 transition. The combined series has also been used in prior work (Mueller, 2025). Province-year fixed effects absorb province-level shifts in reporting around the source transition. Residual changes in relative coverage across cities enter as measurement error in the city-level unrest measure; they would generate the estimated interaction only if they were systematically related to firms' alignment within the same province-year.

Public reporting data from China raise natural concerns about selective reporting and censorship (King et al., 2013). These concerns are less problematic for our empirical design for four reasons. First, prior research has used these sources to study subnational variation in Chinese labor unrest (e.g., Mueller, 2025; Campante et al., 2023; Qin et al., 2019). Second, the Chinese state has often allowed reports of localized labor disputes, in part because such disputes can reveal local grievances and problems in local governance (Campante et al., 2023; Lorentzen, 2014). Third, the CLB series closely matches official data on labor-dispute arbitration in the aggregate (Campante et al., 2023). Finally, our main specifications absorb persistent differences across locations and common changes

over time, reducing the scope for stable geographic differences or aggregate shifts in reporting to drive the results.

D.6 CSRC: Regulatory Penalties

We obtain data on regulatory penalties from a repository of enforcement decisions from the China Securities Regulatory Commission (CSRC). These penalties target firms or personnel for violations of securities-trading regulations and cover 2001–2025. We obtain the universe of decisions from two sections of the CSRC's official website: "Administrative Penalties Decisions" and "Supervision Measures."[48] The former category includes more severe penalties, such as fines and trading bans, while the latter includes softer measures, such as warnings and consultations. We identify and eliminate duplicate listings, yielding a final sample of 1,809 unique cases.

We match CSRC decisions to listed firms by searching for each firm's full and abbreviated names within all decision texts. When a firm's name is found, GPT-4o mini reads the decision text to determine whether the penalty concerns the company itself, its management, or its majority shareholders. Cases unrelated to these three stakeholder groups are excluded. We then collapse the cases to the firm-year level to match the other firm-level observables.

D.7 SAMR: Antitrust Investigations

We obtain data on antitrust decisions issued by the State Administration for Market Regulation (SAMR) over 2013–2025. The dataset is sourced from the SAMR's official website.[49] It contains the title and full text of all decisions. In nearly all cases, the decisions report the date on which the antitrust investigation was initiated and the date on which the final decision was issued. When the initiation date is not separately reported, we use GPT-4o mini to identify the initiation date from the decision text. If this process does not yield an initiation date, we use the issuance date as a proxy. We omit twenty listings that announced investigation terminations. This process yields 409 unique antitrust

[48] http://www.csrc.gov.cn/csrc/c101971/zfxxgk_zdgk.shtml
[49] https://www.samr.gov.cn/fldes/tzgg/xzcf/index.html

cases.

We match the SAMR data to our firm data by searching for each firm's full and abbreviated names within the decision text. We then use GPT-4o mini to verify whether the penalty pertains directly to the firm, its management, or its majority shareholders. Cases not relevant to these groups are omitted from the analysis. We aggregate the data to the firm-year level for subsequent analysis.

The two sources date events differently. We date a SAMR case to the year the investigation opened, or to the issuance year when the text does not report it. CSRC decisions record no initiation date, so those cases carry the year of the enforcement decision.

D.8 Abnormal Returns

We calculate firms' weekly stock returns and abnormal returns using weekly stock-market data for Chinese A-share firms over 2015–2022. We define abnormal returns as

$$AR_{it} = R_{it} - E(R_{it}|X_t),$$

where i indexes stocks and t indexes weeks. R_{it} is stock i's realized return in week t.[50] $E(R_{it}|X_t)$ is the expected return in week t. This measure runs from open to close within the week and therefore excludes any price movement between the previous week's close and the first trading day's open.

Following the index model of MacKinlay (1997), we assume that expected returns are linear in the market return:

$$AR_{it} = R_{it} - (\alpha_i + \beta_i R_{mt}),$$

where R_{mt} is the market return for Chinese A-share firms in week t. For each stock i, we estimate α_i and β_i by regressing R_{it} on R_{mt} using returns from weeks at least four weeks before the event. We calculate abnormal returns for all firms from the first to the last week in the sample.

[50] $R_{it} = \frac{P_{end} - P_{start}}{P_{start}}$, where P_{start} is the stock's opening price on the first trading day of the week and P_{end} is the stock's closing price on the last trading day of the week.

E Other Scandals

In this appendix section, we examine two additional scandals identified through a broader search over civilian officials at the national-deputy rank (*fuguoji*) or above during the period for which we observe stock returns: the public accusation against Zhang Gaoli in November 2021 and the February 2018 sanctioning of Yang Jing. These exercises parallel our analysis of the Sun Zhengcai scandal and test whether firms with greater public alignment experienced lower stock returns around these events than firms with less public alignment.

Zhang Gaoli was a former first-ranked Vice Premier and Politburo Standing Committee member. In November 2021, tennis player Peng Shuai publicly accused Zhang of sexual assault, triggering a major political scandal involving one of the most senior retired officials in China.[51] Yang Jing was a former State Councilor and Secretary-General of the State Council. In February 2018, he was sanctioned for serious violations of Party discipline, effectively ending his political career.[52] These cases differ from Sun Zhengcai's in the nature of political exposure: Zhang was retired at the time of the scandal, and the allegation concerned personal misconduct rather than a governance-related disciplinary matter, while Yang was not a Politburo member. They therefore provide useful tests of whether the relationship between public alignment and stock returns generalizes beyond the baseline event involving an incumbent Politburo member.

For each scandal, we estimate an event study using weekly stock returns of Chinese listed firms in event weeks −2 through +2 around the public disclosure date. The empirical specification is identical to that used in the main text:

$$\text{Return}_{i,t} = \sum_{\substack{k=-2 \\ k\neq -1}}^{2} \beta_k \left(\text{Alignment}_i \times \mathbf{1}\{t = k\}\right) + \zeta_i + \tau_t + \varepsilon_{i,t}. \tag{3}$$

Here, Alignment_i denotes firm i's *PC1* score, measured using the last available annual report before

[51] See CNN, 2021.
[52] See South China Morning Post, 2018.

the relevant event; ζ_i denotes firm fixed effects; and τ_t denotes event-week fixed effects. In the event-study figures, week -1 is omitted as the reference period, so the plotted coefficients trace the evolution of returns relative to the week immediately preceding the scandal. The sample is restricted to the $[-2, 2]$ event window. Both the outcome and the alignment measure are standardized within this locked sample. Standard errors are clustered at the firm level.

Figure A.7(c) presents the estimates for the Zhang Gaoli scandal. The coefficients in the event week and the following week are -0.115 and -0.079, respectively, indicating lower returns among more-aligned firms relative to less-aligned firms. However, the week -2 coefficient is 0.077 and statistically significant at the 1% level. Because week -1 is the omitted period, this estimate indicates a significant pre-event differential in returns by alignment.

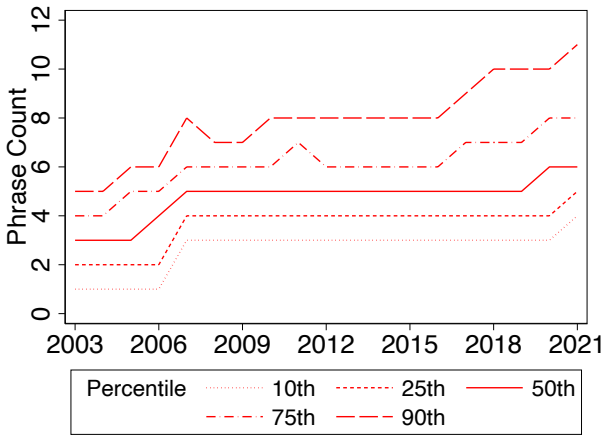
Figure A.7(d) presents the corresponding estimates for the Yang Jing scandal. The coefficients in the event week and the following week are -0.093 and -0.091, respectively, and both are statistically significant. The week -2 coefficient is also negative, at -0.040, and statistically significant. Thus, more-aligned firms experienced lower differential returns in week -2 relative to the omitted week -1.

Taken together, both supplemental events exhibit negative event-period coefficients, but both also contain statistically significant pre-event differences between more- and less-aligned firms. Moreover, the week -2 coefficients have opposite signs across the two events. We therefore interpret them as suggestive evidence that negative event-period returns among more-aligned firms are not unique to the Sun Zhengcai episode, but avoid a sharp causal interpretation due to pre-period differences in week -2.

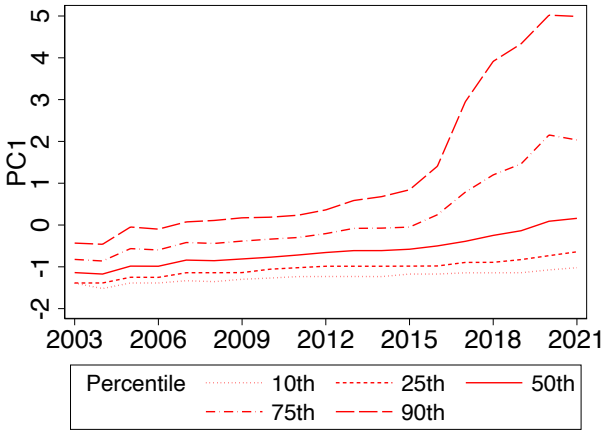
Appendix Tables and Figures

Figure A.6: Public Alignment Over Time

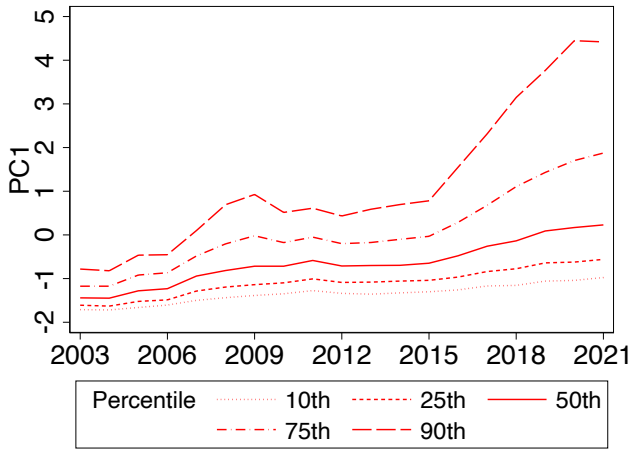
(a) Phrase Count, First Text Portion



(b) PC1, First Text Portion



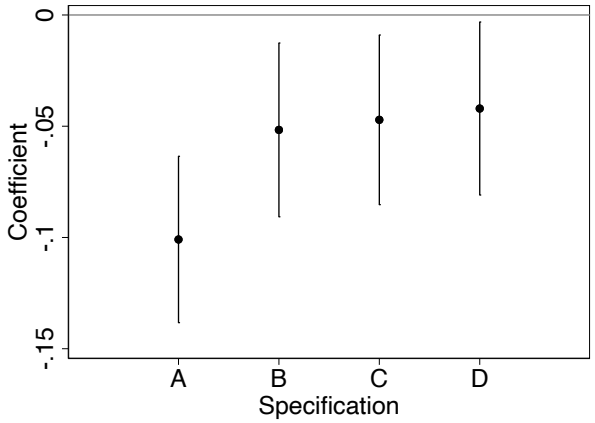
(c) PC1 of Full Text Phrase Counts



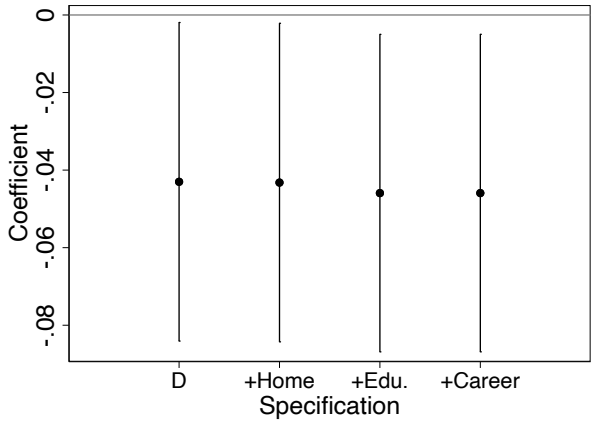
Notes: This figure shows the distribution of public alignment across listed Chinese firm-years from 2003 to 2021. Annual reports come from WIND. The 50 Party-related phrases are identified using TF-IDF analysis of CCP documents. Figures A.6(a) and A.6(b) use the first 30,689 words of each report, roughly one standard deviation of average report length, and record whether each phrase appears. Figure A.6(a) shows the number of distinct phrases mentioned. Figure A.6(b) shows the first principal component of the phrase indicators. Figure A.6(c) uses full-report phrase counts and shows the first principal component of these counts.

Figure A.7: Stock Returns Around Political Scandals, Robustness

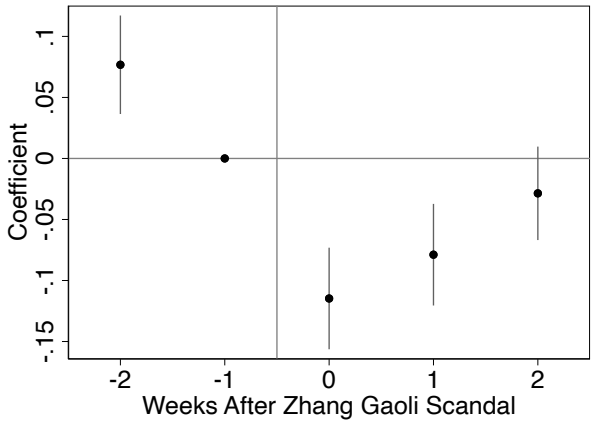
(a) Abnormal Returns



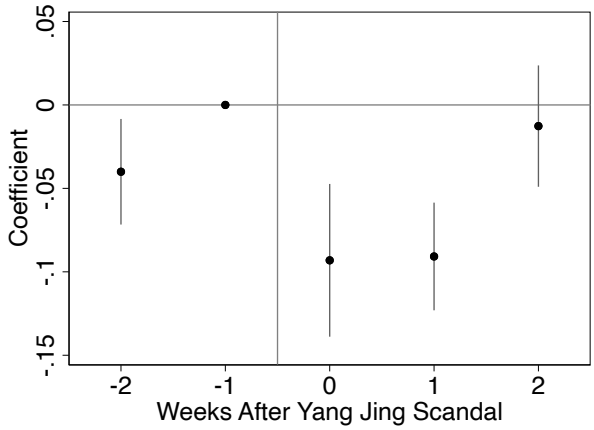
(b) Controls for Geographic Links to Sun Zhengcai



(c) Zhang Gaoli Scandal



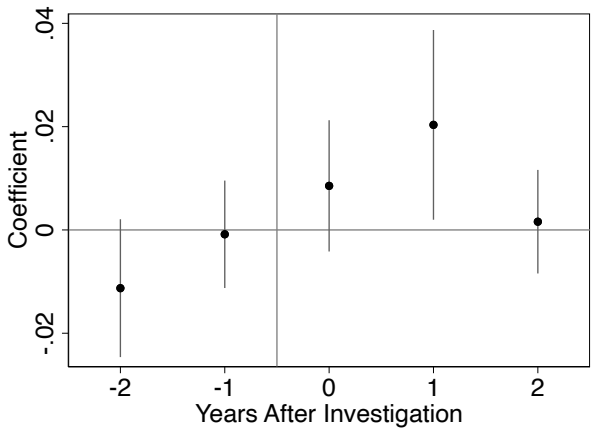
(d) Yang Jing Scandal



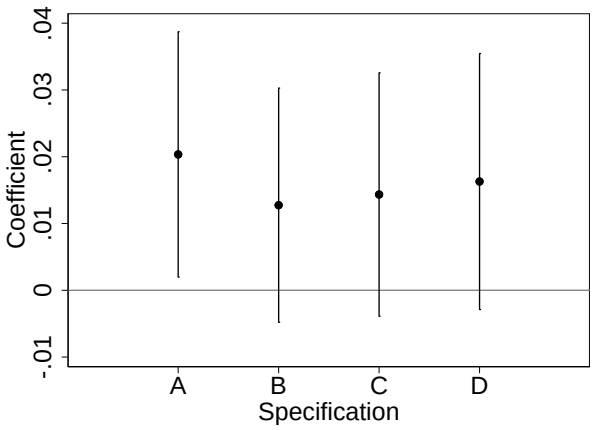
Notes: The unit of observation is a firm-week, with event windows from two weeks before through two weeks after each scandal. Returns are constructed from daily CSMAR data. Figures A.7(a) and A.7(b) use the Sun Zhengcai scandal: Figure A.7(a) uses abnormal returns and progressively adds interactions between Post and SOE status, board political connections, and Party-cell adoption; Figure A.7(b) uses weekly raw returns and further adds interactions between Post and indicators for headquarters, schooling, and prior-office links to Sun. Figures A.7(c) and A.7(d) show weekly return responses around the Zhang Gaoli and Yang Jing scandals, using alignment-by-week interactions with week -1 omitted. Regressions include firm and event-week fixed effects. Standard errors are clustered by firm. Bars show 95% confidence intervals.

Figure A.8: Public Alignment Around Regulatory Investigations

(a) Event Study

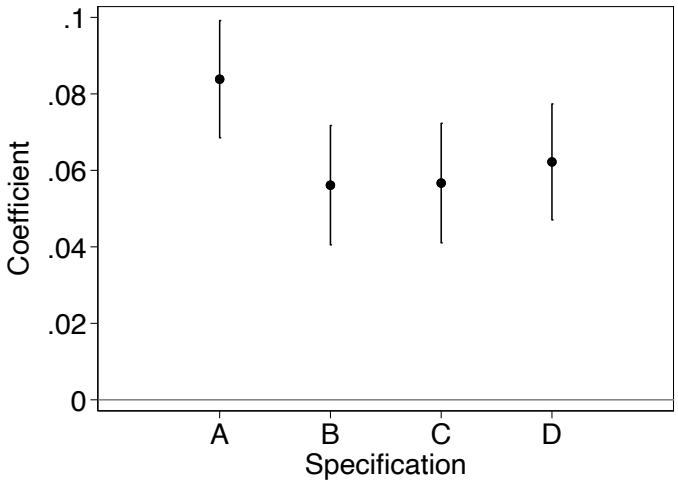


(b) Robustness to Closeness to the State



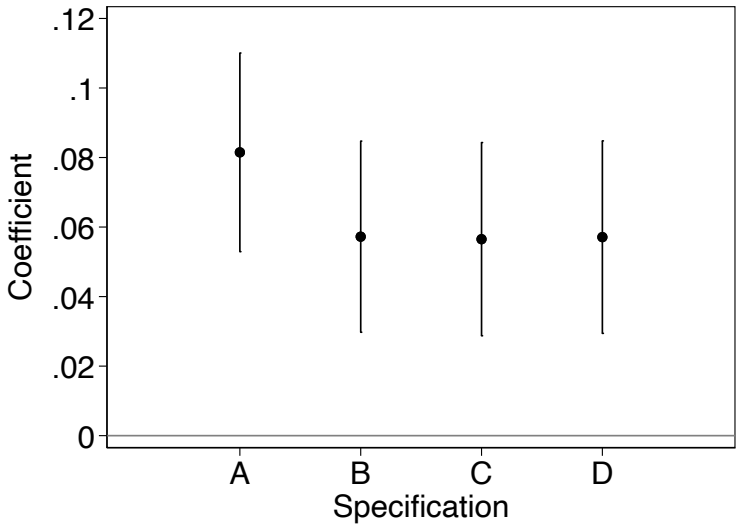
Notes: The unit of observation is a firm-year. Investigations are measured as the number of SAMR antitrust cases and CSRC regulatory penalties of which the firm is a subject in the year. The outcome is Public Alignment. Figure A.8(a) is an event study that plots coefficients on investigation counts from two years ahead through two years prior; the horizontal axis therefore measures time relative to the investigation, with negative values denoting years before it. Figure A.8(b) plots the one-year-post-investigation coefficient as the specification progressively adds, for each of SOE status, board political connections, and Party-cell adoption, year interactions together with interactions with every investigation lead and lag, so the response to regulatory scrutiny may differ by each characteristic. The interacted characteristics are demeaned, so the plotted coefficient is the average response. The estimation sample is held fixed at the most saturated specification. All regressions include firm and year fixed effects. Standard errors are clustered by firm. Bars show 95% confidence intervals.

Figure A.9: CSR and Public Alignment, Robustness to Closeness to the State



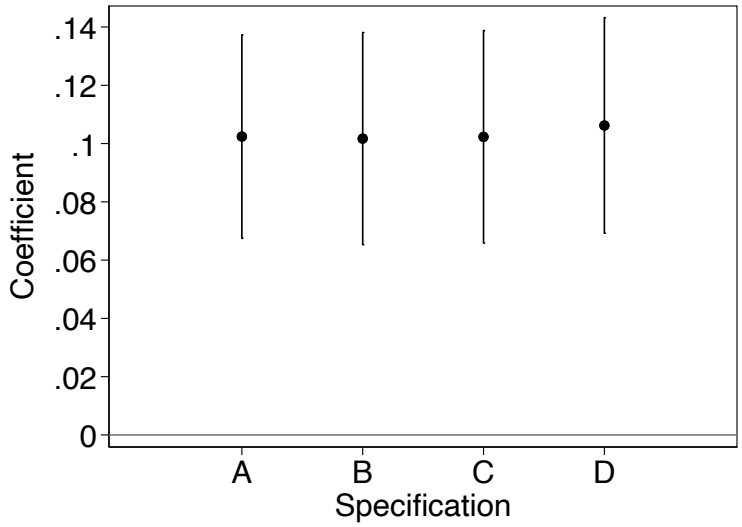
Notes: The unit of observation is a firm-year. The figure plots the coefficient from a regression of public alignment on the firm's overall CSR score. The public-alignment score and CSR score are standardized to have mean zero and standard deviation one within the estimation sample. Coefficients are reported across four specifications: (A) year fixed effects; (B) additionally ownership fixed effects; (C) additionally political-connection fixed effects; and (D) additionally Party-cell fixed effects. The estimation sample is held fixed at the most saturated specification. Standard errors are clustered by firm. Bars represent 95% confidence intervals.

Figure A.10: Subsidies and Public Alignment



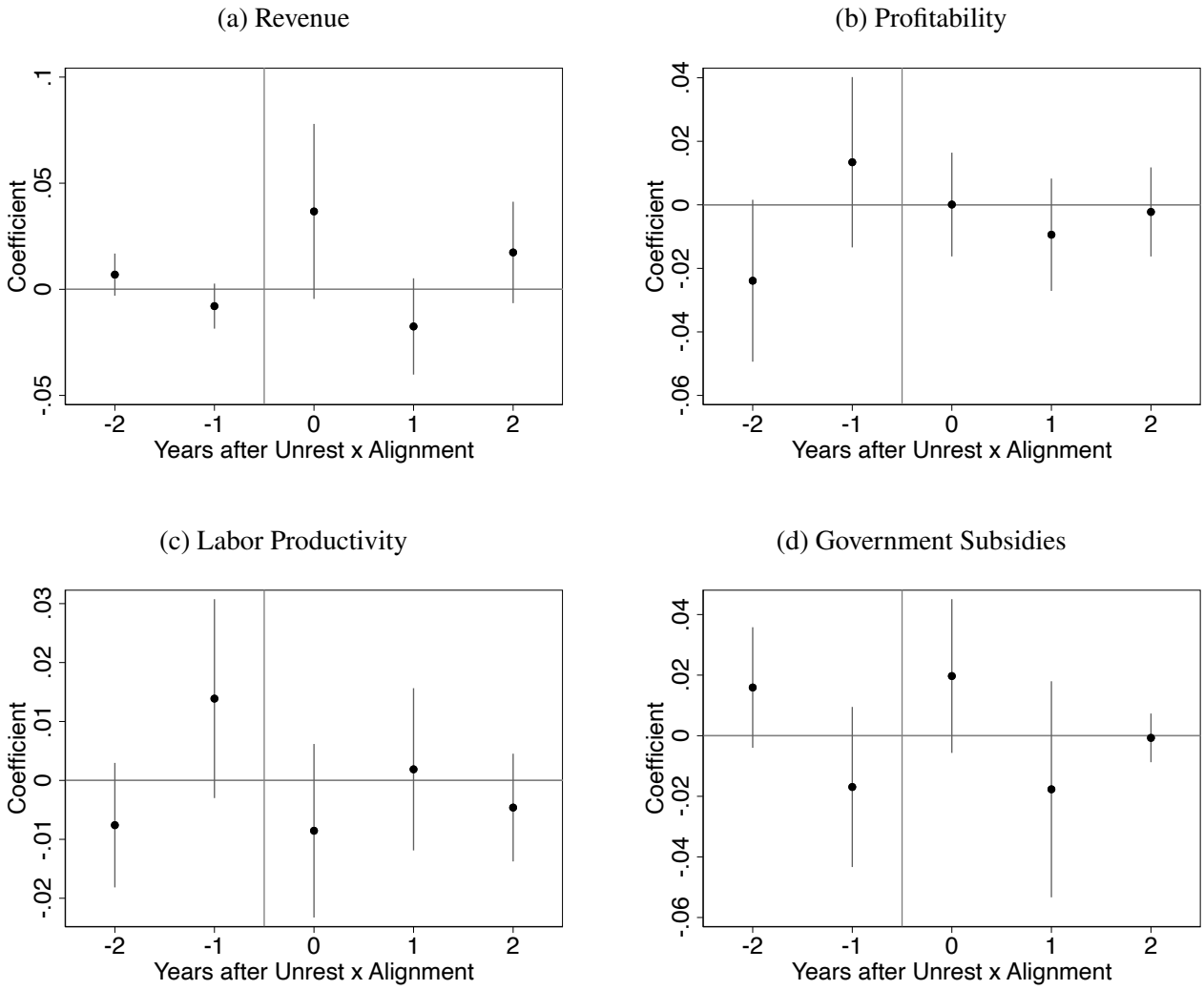
Notes: The unit of observation is a firm-year. The figure plots the coefficient on lagged public alignment, measured as the first principal component of Party-related phrase indicators. The dependent variable is total government subsidies received by the firm. The dependent variable and the public-alignment score are standardized to have mean zero and standard deviation one within the estimation sample. Coefficients are reported across four specifications: (A) year fixed effects; (B) additionally ownership fixed effects; (C) additionally political-connection fixed effects; and (D) additionally Party-cell fixed effects. The estimation sample is held fixed at the most saturated specification. Standard errors are clustered by firm. Bars represent 95% confidence intervals.

Figure A.11: Government Procurement and Public Alignment



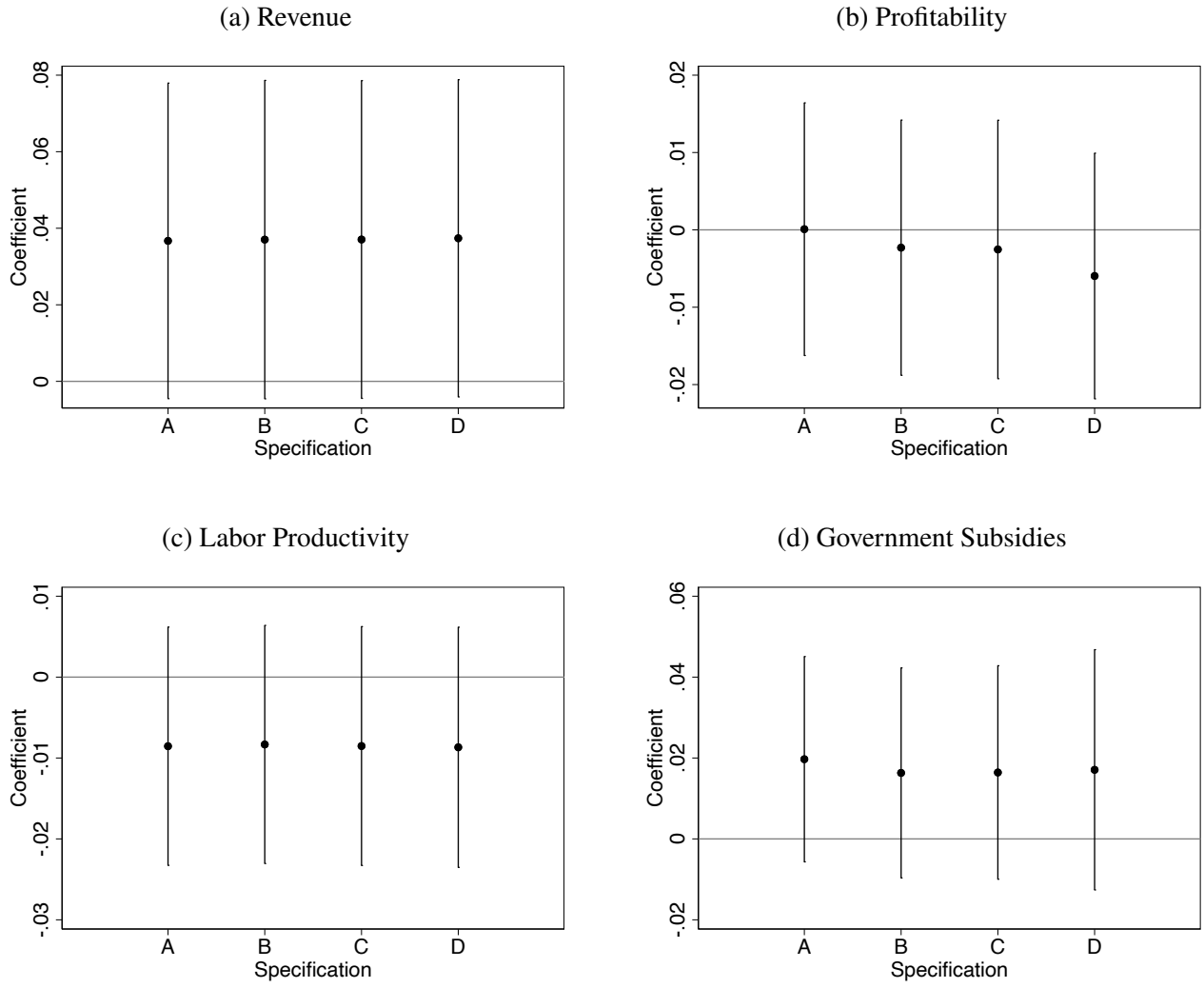
Notes: The unit of observation is a firm-year. The figure plots the coefficient on lagged public alignment, measured as the first principal component of Party-related phrase indicators. The dependent variable is the number of government procurement contracts awarded to the firm that year. The dependent variable and the public-alignment score are standardized to have mean zero and standard deviation one within the estimation sample. Coefficients are reported across four specifications: (A) year fixed effects; (B) additionally ownership fixed effects; (C) additionally political-connection fixed effects; and (D) additionally Party-cell fixed effects. The estimation sample is held fixed at the most saturated specification. Standard errors are clustered by firm. Bars represent 95% confidence intervals.

Figure A.12: Firm Performance Response to Unrest by Public Alignment



Notes: The unit of observation is a firm-year. Each panel presents a distributed lead-and-lag specification, plotting coefficients on the interaction between city-level labor unrest and firm-level public alignment, measured as the first principal component of Party-related phrase indicators, by event time. Negative event times denote years before unrest, and positive event times denote years after unrest. All dependent variables, the public-alignment score, and the unrest variables are standardized to have mean zero and standard deviation one within the estimation sample. The specification controls for uninteracted unrest at all leads and lags, alignment, firm fixed effects, and province-year fixed effects, and is estimated on the sample of the most saturated specification of Table A.11. Standard errors are clustered at the city level. Bars represent 95% confidence intervals.

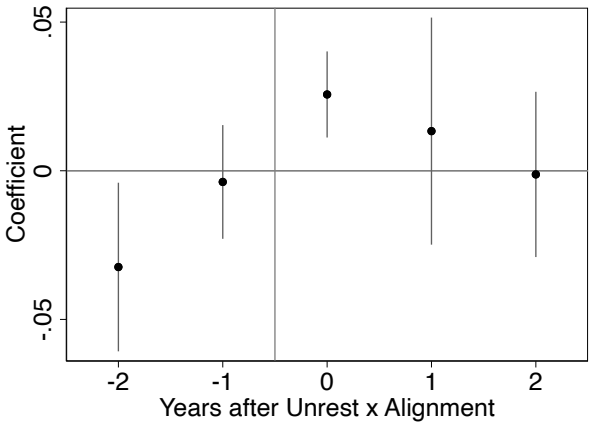
Figure A.13: Firm Performance Response to Unrest by Public Alignment, Robustness to Closeness to the State



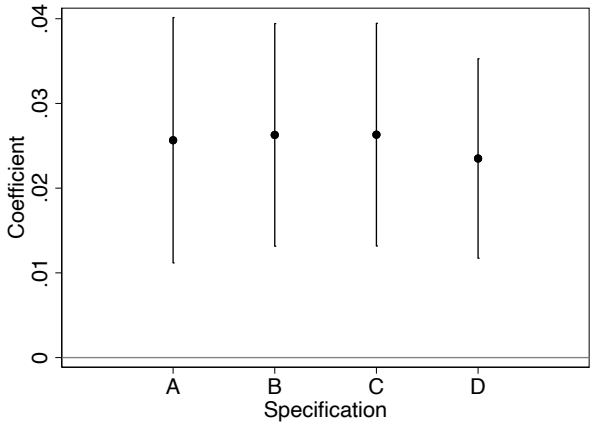
Notes: The unit of observation is a firm-year. Each panel plots coefficients on the contemporaneous interaction between unrest and alignment across four specifications: (A) the baseline with uninteracted unrest at all leads and lags, alignment, firm fixed effects, and province-year fixed effects; (B) additionally interacting unrest at all leads and lags with ownership fixed effects; (C) additionally interacting unrest with political connection fixed effects; and (D) additionally interacting unrest with Party-cell fixed effects. Dependent variables are defined as in Figure A.12. Standard errors are clustered at the city level. Bars represent 95% confidence intervals.

Figure A.14: Procurement Response to Unrest by Public Alignment

(a) Baseline



(b) Controlling for Closeness to the State



Notes: The unit of observation is a firm-year. The dependent variable is the number of government procurement contracts awarded to the firm that year, standardized to have mean zero and standard deviation one within the estimation sample, as are the public-alignment score and the unrest variables. Figure A.14(a) presents a distributed-lag model, plotting coefficients on the interaction between city-level labor unrest and firm-level public alignment across different leads and lags, controlling for uninteracted unrest at all leads and lags, alignment, firm fixed effects, and province-year fixed effects. Figure A.14(b) reports coefficients on unrest interacted with alignment across four specifications: (A) the baseline; (B) additionally interacting unrest at all leads and lags with ownership fixed effects; (C) additionally interacting unrest with political connection fixed effects; and (D) additionally interacting unrest with Party-cell fixed effects. Standard errors are clustered at the city level. Bars represent 95% confidence intervals.

Table A.2: Theoretical Objects, Predicted Patterns, and Empirical Evidence

Object	Theoretical implication	Empirical pattern	Corresponding evidence
Mean payoff response	No differential mean payoff responses absent screening; in optimal screening, more-aligned firms are more exposed.	Differential mean payoff responses rule in screening; larger losses in more-aligned firms are consistent with optimal screening.	More-aligned firms experience larger stock-price declines after the Sun Zhengcai scandal.
Conditional support, x_a, and *ex ante* support burden, $s_a = qx_a$	No differential support burdens absent screening; in optimal screening, more-aligned firms have higher support burdens: $s_1^* = q > s_0^* = 0$.	Differential support burdens rule in a screening role; more-aligned firms should provide more costly support when political stress materializes under optimal screening.	More-aligned firms increase employment more during local labor unrest. They have higher CSR scores, consistent with undertaking other costly regime-oriented activities.
Opportunity cost of support, c and $\mathbf{E}[c]$	When both contracts are chosen, aligned firms have lower costs. Alignment probability falls with upward shifts in costs.	Alignment should be more likely when support is less costly or when firms' opportunity costs of support are lower.	Firms increase alignment after regulatory investigations, consistent with investigations lowering firms' opportunity costs of support.
Transfer or favorable treatment, t_a	When both contracts are chosen, the aligned contract carries a higher transfer: $t_1^* > t_0^*$.	More-aligned firms should receive more favorable treatment.	Higher public alignment is associated with more government subsidies and more government procurement contracts.
Cutoff type, $c^* = V_1 + \frac{U_1 - U_0}{q}$	The probability of alignment is nondecreasing in c^*. The value of screening is maximized at $c^* = \mathbf{E}[c]$, and may fall as alignment becomes more prevalent.	An increase in c^* makes alignment more prevalent, but prevalence alone does not identify the value of screening.	Alignment rises over time, especially in the upper tail, consistent with alignment becoming more attractive.
Dispersion of support costs, F	The value of screening is nondecreasing under mean-preserving spreads of F.	Screening is more valuable when firms' support costs are more dispersed.	The decomposition of alignment shows substantial firm-level and firm-year variation, consistent with heterogeneity in firms' costs and constraints.

Notes: The table summarizes the theoretical implications of Propositions 1 to 4 and relates them to the empirical evidence. The cutoff c^* is the highest cost type that aligns under the surplus-maximizing menu. The first two rows summarize the tests that rule in a screening role and assess whether the direction of the responses is consistent with optimal screening. The remaining rows report additional patterns consistent with optimal screening; they do not identify the primitives of the theoretical framework or directly test every comparative static.

Table A.3: Decomposition of Variation in Public Alignment

	Adjusted R-squared	Difference in adjusted R-squared from Year FE	Observations
+ Year FE	0.184		45,061
+ Industry FE	0.221	0.037	45,061
+ City FE	0.253	0.069	45,061
+ Industry-by-City FE	0.294	0.110	45,061
+ Industry-by-City-by-Year FE	0.287	0.103	45,061
+ Firm FE	0.506	0.322	45,061

Notes: This table decomposes the main sources of variation in our baseline measure of alignment. The dependent variable is the firm's public alignment, measured as the first principal component of Party-related phrase indicators. Each row adds the listed fixed effects cumulatively to all rows above, and all rows are estimated on the same sample. We report the adjusted R^2, which accounts for the number of absorbed fixed effects; this matters for the finer interactions, as 7,081 of the 12,651 industry-by-city-by-year cells contain a single firm-year. The third column reports the difference between each row's adjusted R^2 and that of the year-fixed-effects row.

Table A.4: Variable Definitions, Sources, and Units

Variable	Description	Source	Unit / Type
Public Alignment	The first principal-component and phrase-count measures from a vector of indicators for whether 50 phrases related to CCP regime speech occur in firm i's annual report in year t.	Phrases are identified from a corpus of regime speech using TF-IDF. The baseline corpus includes State Council circulars and General Office circulars, Central Committee plenary reports, and *People's Daily* articles.	Principal component / count
Scandal	Indicates the event week of a political scandal in the event-study window. The baseline event is the Sun Zhengcai scandal (2017); robustness analyses use the Zhang Gaoli (2021) and Yang Jing (2018) scandals.	Coding by the authors as described in the paper.	Dummy
Unrest	Number of labor-unrest events in firm i's home city in year t.	Elfstrom (2017) for 2003–2011 and China Labour Bulletin (2019) for 2012–2022.	Count
Employees	Total number of employees of firm i in year t.	Shenzhen CSMAR Data Technology Co., Ltd. (2024), 2003–2019.	Count
SAMR Case	Number of investigations or cases by the State Administration for Market Regulation (SAMR) of which firm i is a subject in year t.	Scraped by the authors from the official SAMR website (State Administration for Market Regulation, 2025).	Count
CSRC Case	Number of investigations or cases by the China Securities Regulatory Commission (CSRC) of which firm i is a subject in year t.	Scraped by the authors from the official CSRC website (China Securities Regulatory Commission, 2025).	Count
Subsidies	Total government subsidies received by firm i in year t, measured from the government-grants disclosure schedule of the annual report; when the disclosure schedule is missing, the sum of the other-income and non-operating-income government-subsidy line items is used instead.	Wind Information Co., Ltd. (2025), 2007–2021.	Continuous
Procurement	Number of government procurement contracts awarded to firm i in year t, based on matching firm names to all published tender-award announcements and classifying the firm's role in each contract.	Collected by the authors from the China Government Procurement network (CCGP), 2013–2020.	Count
Ownership	Indicates whether firm i is state-owned in year t.	Shenzhen CSMAR Data Technology Co., Ltd. (2024).	Dummy
Political Connection	Indicates whether firm i had at least one politically connected director before Regulation No. 18 in October 2013, which banned politically connected directors from the boards of listed firms. Politically connected directors include CPC and government officials, National People's Congress deputies, Chinese People's Political Consultative Conference representatives, and leaders of state-owned enterprises and public or non-profit institutions.	Fan (2021).	Dummy
Party Cell	Indicates whether firm i has adopted a corporate-charter provision establishing a Party cell by year t; once the indicator turns on, it remains equal to one in subsequent years.	Charter amendments from Wind Information Co., Ltd. (2025); text search for "Party cell" over 2003–2021.	Dummy
Political Risk	Firm-level political risk of firm i in year t, constructed from the annual report following Hassan et al. (2019): the distinctiveness-weighted share of political-dictionary words that appear within ten words of a risk or uncertainty term. See Appendix C.2.	Calculation by the authors based on the same annual reports used for Public Alignment.	Continuous
Sector	Firm i's primary industry sector, defined using the first-level industry code in WIND.	Wind Information Co., Ltd. (2025), first-level industry classification.	Categorical

Variable	Description	Source	Unit / Type
Sales	Firm i's annual sales, measured as operating revenue, in year t. Summary statistics report the unwinsorized series. Regressions use sales_{w2}, winsorized at the 0.1 and 99.9 percentiles.	Shenzhen CSMAR Data Technology Co., Ltd. (2024), 2004–2021.	Continuous
Profitability	Firm i's profitability in year t, defined as profitability = $\text{profit}_{w2}/\text{sales}_{w2}$, where both profit_{w2} and sales_{w2} are winsorized at the 0.1 and 99.9 percentiles.	Own calculation based on Shenzhen CSMAR Data Technology Co., Ltd. (2024), 2004–2021.	Continuous
Labor Productivity	Firm i's labor productivity in year t, defined as $\text{sales}_{w2}/\text{employees}$, where sales_{w2} is winsorized at the 0.1 and 99.9 percentiles.	Own calculation based on Shenzhen CSMAR Data Technology Co., Ltd. (2024).	Continuous
CSR score	Firm i's CSR score in year t.	CnOpenData (2025), 2010–2020.	Continuous
Weekly return	Simple stock return for firm i in calendar week w, calculated as $(P_{iw}^{\text{close}} - P_{iw}^{\text{open}})/P_{iw}^{\text{open}}$, where the opening price is from the first trading day of the week and the closing price is from the last trading day of the week.	Shenzhen CSMAR Data Technology Co., Ltd. (2024) daily stock prices.	Continuous
Abnormal return	Weekly abnormal stock return for firm i in week t, equal to the realized weekly return minus the expected return from an index (market) model, $AR_{it} = R_{it} - (\alpha_i + \beta_i R_{mt})$, following MacKinlay (1997). See Appendix D.8.	Calculation by the authors based on weekly returns from Shenzhen CSMAR Data Technology Co., Ltd. (2024).	Continuous

Table A.5: Summary Statistics

Variable	Mean	SD	N	Min	Max	Year coverage
Panel A: Firm Characteristics						
Public Alignment	-0.000	1.887	45,061	-2.590	15.088	2003–2021
Unrest	13.215	18.224	45,061	0	100	2003–2021
Employees (Thousands)	5.974	23.813	37,348	0	553	2003–2019
SAMR Case	0.014	0.394	45,061	0	30	2003–2021
CSRC Case	0.103	1.534	45,061	0	109	2003–2021
Subsidies (Million RMB)	43.554	331.15	35,941	-23.461	50,342	2007–2021
Procurement Contracts	1.840	11.828	28,702	0	401	2013–2020
Ownership	0.440	0.496	37,646	0	1	2003–2020
Political Connection	0.331	0.471	45,061	0	1	2003–2021
Party Cell	0.137	0.344	45,061	0	1	2003–2021
Sales (Billion RMB)	9.514	74.027	43,365	0	3,220.2	2004–2021
Profitability	0.077	0.272	43,142	-1.585	0.869	2004–2021
CSR score	23.682	15.861	31,559	-18.450	90.870	2010–2020
Panel B: Market Returns						
Weekly return	0.002	0.072	1,423,830	-0.896	19.769	2015–2022
Abnormal return	-0.001	0.067	1,423,814	-1.181	19.769	2015–2022

Notes: Panel A presents firm-year-level characteristics. Employees is measured in thousands, Sales is measured in billion RMB, and Subsidies is measured in million RMB. Procurement Contracts is the number of government procurement contracts awarded to the firm that year. Ownership, Political Connection, and Party Cell are indicator variables equal to one if the respective condition is met, and zero otherwise. SAMR Case and CSRC Case refer to cases involving the State Administration for Market Regulation and the China Securities Regulatory Commission, respectively. Panel B reports firm-week-level market-return statistics, where Abnormal return denotes the weekly abnormal return, computed from an index (market) model as described in Appendix D.8.

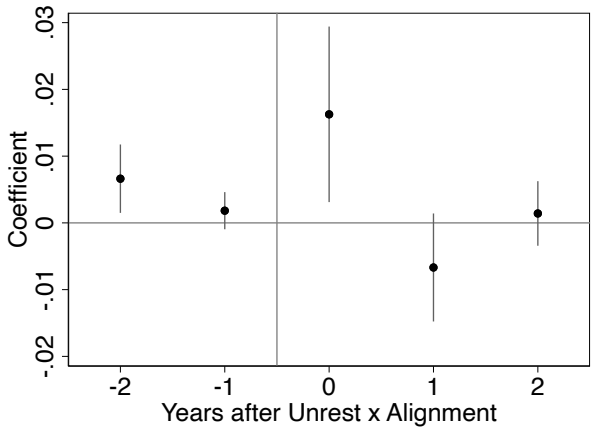
Table A.6: Employment Response to Unrest by Public Alignment, Robustness to Closeness to the State

	(1)	(2)	(3)	(4)
	# Employees	# Employees	# Employees	# Employees
$\text{Alignment}_{i,t} \times \text{Unrest}_{c(i),t+2}$	0.007**	0.008**	0.007**	0.007**
	(0.003)	(0.003)	(0.003)	(0.003)
$\text{Alignment}_{i,t} \times \text{Unrest}_{c(i),t+1}$	0.003*	0.001	0.001	-0.000
	(0.001)	(0.001)	(0.001)	(0.001)
$\text{Alignment}_{i,t} \times \text{Unrest}_{c(i),t}$	0.016**	0.016**	0.016**	0.017**
	(0.007)	(0.007)	(0.007)	(0.007)
$\text{Alignment}_{i,t} \times \text{Unrest}_{c(i),t-1}$	-0.006	-0.006	-0.006	-0.007
	(0.004)	(0.004)	(0.004)	(0.005)
$\text{Alignment}_{i,t} \times \text{Unrest}_{c(i),t-2}$	0.001	0.002	0.002	0.002
	(0.002)	(0.002)	(0.002)	(0.002)
Observations	34,753	34,753	34,753	34,753
R-squared	0.954	0.954	0.954	0.954
Firm FE	Yes	Yes	Yes	Yes
Year × Province FE	Yes	Yes	Yes	Yes
Unrest × SOE FE	No	Yes	Yes	Yes
Unrest × Political Connection FE	No	No	Yes	Yes
Unrest × Party cell FE	No	No	No	Yes

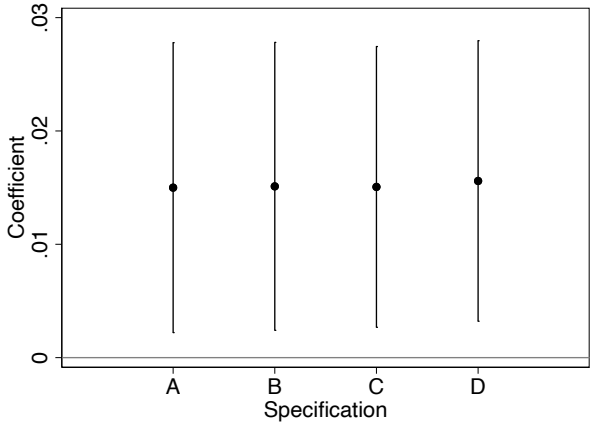
Notes: The unit of observation is a firm-year. The dependent variable is the total number of employees. The independent variables of interest are interactions between the firm's public alignment, measured as the first principal component of Party-related phrase indicators, and labor unrest in the firm's city from $t-2$ through $t+2$. Because employment is measured in year t, the $t+1$ and $t+2$ interactions are the one- and two-year pre-period coefficients, whereas the $t-1$ and $t-2$ interactions are the one- and two-year post-unrest coefficients. To facilitate interpretation, the dependent variable, the public-alignment score, and the unrest variables are standardized to have mean zero and standard deviation one within the estimation sample. Results are reported across four specifications. Specification (1) controls for uninteracted unrest at all leads and lags, alignment, firm fixed effects, and province-year fixed effects. Specification (2) additionally interacts every unrest lead and lag with state ownership. Specifications (3) and (4) additionally interact every unrest lead and lag with political connections and Party-cell adoption, respectively. Standard errors are clustered at the city level. Significance levels are indicated by * $p < 0.1$, ** $p < 0.05$, and *** $p < 0.01$.

Figure A.15: Employment Response to Unrest by Public Alignment, Controlling for Political Risk

(a) Baseline

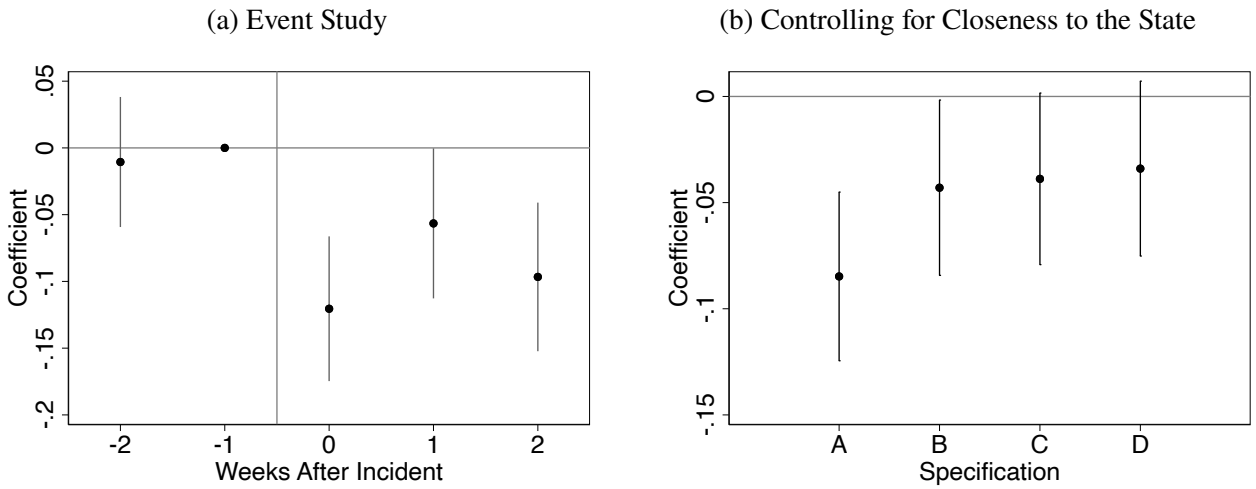


(b) Controlling for Closeness to the State



Notes: This figure reproduces Figure 4(a) while additionally controlling for a firm-level political-risk measure constructed from the same annual reports by replicating Hassan et al. (2019) (Appendix C.2). Political risk enters in exactly the same way as public alignment: as a main effect and interacted with unrest at every lead and lag. The unit of observation is a firm-year; the outcome is the number of employees. Figure A.15(a) plots the alignment–unrest coefficients by event time, with negative event times denoting years before unrest and positive event times denoting years after unrest, while controlling for the analogous unrest × political-risk interactions. Figure A.15(b) plots the contemporaneous $\text{Alignment}_{it} \times \text{Unrest}_{c(i)t}$ coefficient across specifications A–D, which progressively add interactions of unrest with SOE status, political connections, and Party-cell adoption. All specifications control for uninteracted unrest at all leads and lags, alignment, political risk, firm fixed effects, and province-year fixed effects. Standard errors are clustered at the city level. Bars show 95% confidence intervals.

Figure A.16: Firm Valuation After the Sun Zhengcai Scandal, Controlling for Political Risk



Notes: This figure reproduces the Sun Zhengcai event study of Figure 3(a) while additionally controlling for a firm-level political-risk measure constructed from the same annual reports by replicating Hassan et al. (2019) (Appendix C.2). Political risk enters in exactly the same way as public alignment. The unit of observation is a firm-week; the outcome is the weekly stock return, standardized within the $[-2, 2]$ event window. Figure A.16(a) plots coefficients on interactions between firm-level alignment and event-week indicators, with week -1 omitted, controlling for the analogous political-risk $\times$ event-week interactions. Figure A.16(b) plots the coefficient on $\text{Post}_t \times \text{Alignment}_i$ across specifications A–D, controlling for $\text{Post}_t \times \text{Political Risk}_i$; specification A includes firm and event-week fixed effects, and B–D progressively add interactions of Post_t with SOE status, political connections, and Party-cell adoption. Standard errors are clustered at the firm level. Bars show 95% confidence intervals.

Table A.7: Firm Valuation After the Sun Zhengcai Scandal, Controlling for Political Risk

	(1) Weekly return	(2) Weekly return	(3) Weekly return	(4) Weekly return
$\text{Post}_t \times \text{Alignment}_{i,t}$	-0.0848***	-0.0430**	-0.0388*	-0.0340
	(0.0203)	(0.0211)	(0.0206)	(0.0210)
$\text{Post}_t \times \text{Political Risk}_{i,t}$	-0.110***	-0.0757***	-0.0741***	-0.0731***
	(0.0183)	(0.0181)	(0.0178)	(0.0179)
Observations	15,254	15,254	15,254	15,254
R-squared	0.268	0.276	0.280	0.281
Firm FE	Yes	Yes	Yes	Yes
Event-week FE	Yes	Yes	Yes	Yes
Post × SOE FE	No	Yes	Yes	Yes
Post × Political Connection FE	No	No	Yes	Yes
Post × party cell FE	No	No	No	Yes

Notes: This table reports the weekly stock-return response to the Sun Zhengcai scandal by public alignment, controlling for a firm-level political-risk measure constructed by replicating Hassan et al. (2019) (Appendix C.2), which enters in exactly the same way as public alignment. The unit of observation is a firm-week; the outcome is the weekly stock return, standardized within the $[-2, 2]$ event window. Post_t equals one for the event week and subsequent weeks. Specification (1) includes firm and event-week fixed effects. Specifications (2)–(4) progressively add interactions of Post_t with SOE status, political connections, and Party-cell adoption. Both the outcome and the regressors are standardized within the estimation sample. Standard errors are clustered at the firm level. Significance levels are indicated by * p<0.10, ** p<0.05, *** p<0.01.

Table A.8: Employment Response to Unrest by Public Alignment, Controlling for Political Risk

	(1) # Employees	(2) # Employees	(3) # Employees	(4) # Employees
$\text{Alignment}_{i,t} \times \text{Unrest}_{c(i),t+2}$	0.007**	0.007**	0.007**	0.007**
	(0.003)	(0.003)	(0.003)	(0.003)
$\text{Alignment}_{i,t} \times \text{Unrest}_{c(i),t+1}$	0.001	-0.000	-0.000	-0.001
	(0.001)	(0.001)	(0.001)	(0.001)
$\text{Alignment}_{i,t} \times \text{Unrest}_{c(i),t}$	0.015**	0.015**	0.015**	0.016**
	(0.006)	(0.006)	(0.006)	(0.006)
$\text{Alignment}_{i,t} \times \text{Unrest}_{c(i),t-1}$	-0.007**	-0.007**	-0.007**	-0.008**
	(0.003)	(0.003)	(0.003)	(0.003)
$\text{Alignment}_{i,t} \times \text{Unrest}_{c(i),t-2}$	0.001	0.001	0.001	0.002
	(0.002)	(0.002)	(0.002)	(0.003)
$\text{Political Risk}_{i,t} \times \text{Unrest}_{c(i),t+2}$	0.003	0.003*	0.003**	0.003*
	(0.002)	(0.002)	(0.002)	(0.002)
$\text{Political Risk}_{i,t} \times \text{Unrest}_{c(i),t+1}$	0.002	0.001	0.001	0.001
	(0.002)	(0.002)	(0.002)	(0.002)
$\text{Political Risk}_{i,t} \times \text{Unrest}_{c(i),t}$	0.002	0.003	0.003	0.003
	(0.003)	(0.002)	(0.002)	(0.002)
$\text{Political Risk}_{i,t} \times \text{Unrest}_{c(i),t-1}$	0.002	0.002	0.002	0.002
	(0.002)	(0.002)	(0.002)	(0.002)
$\text{Political Risk}_{i,t} \times \text{Unrest}_{c(i),t-2}$	-0.002	-0.001	-0.001	-0.001
	(0.003)	(0.003)	(0.003)	(0.003)
Observations	29,422	29,422	29,422	29,422
R-squared	0.956	0.956	0.956	0.956
Firm FE	Yes	Yes	Yes	Yes
Year × Province FE	Yes	Yes	Yes	Yes
Unrest × SOE FE	No	Yes	Yes	Yes
Unrest × Political Connection FE	No	No	Yes	Yes
Unrest × Party cell FE	No	No	No	Yes

Notes: This table reports the employment response to local unrest by public alignment, controlling for a firm-level political-risk measure constructed by replicating Hassan et al. (2019) (Appendix C.2), which enters in exactly the same way as public alignment: as a main effect and interacted with unrest at every lead and lag. The unit of observation is a firm-year; the dependent variable is the number of employees. Because employment is measured in year t, interactions with unrest at $t+1$ and $t+2$ are the one- and two-year pre-period coefficients, whereas interactions with unrest at $t-1$ and $t-2$ are the one- and two-year post-unrest coefficients. The dependent variable, alignment, political risk, and the unrest variables are standardized within the estimation sample. Specification 1 controls for uninteracted unrest at all leads and lags, alignment, political risk, firm fixed effects, and province-year fixed effects. Specifications 2–4 progressively add interactions of unrest with SOE status, political connections, and Party-cell adoption. Standard errors are clustered at the city level. Significance levels are indicated by * p<0.10, ** p<0.05, *** p<0.01.

Table A.9: Subsidies and Lagged Public Alignment

	(1) Subsidy	(2) Subsidy	(3) Subsidy	(4) Subsidy
$\text{Alignment}_{i,t-1}$	0.081***	0.057***	0.057***	0.057***
	(0.015)	(0.014)	(0.014)	(0.014)
Observations	26,639	26,639	26,639	26,639
R-squared	0.011	0.024	0.025	0.025
Year FE	Yes	Yes	Yes	Yes
SOE FE	No	Yes	Yes	Yes
Political Connection FE	No	No	Yes	Yes
Party cell FE	No	No	No	Yes

Notes: The unit of observation is a firm-year. The dependent variable is total government subsidies received by the firm, winsorized at the 0.1 and 99.9 percentiles. The independent variable is the firm's public alignment in the previous year, measured as the first principal component of Party-related phrase indicators. Both are standardized to have mean zero and standard deviation one within the estimation sample. Specification 1 includes year fixed effects. Specification 2 adds ownership fixed effects. Specification 3 adds political-connection fixed effects. Specification 4 adds Party-cell fixed effects. The estimation sample is held fixed at the most saturated specification. Standard errors are clustered by firm. Significance levels are indicated by * p<0.10, ** p<0.05, *** p<0.01.

Table A.10: Government Procurement and Lagged Public Alignment

	(1) Procurement	(2) Procurement	(3) Procurement	(4) Procurement
$\text{Alignment}_{i,t-1}$	0.102***	0.102***	0.102***	0.106***
	(0.018)	(0.019)	(0.019)	(0.019)
Observations	20,006	20,006	20,006	20,006
R-squared	0.011	0.011	0.011	0.012
Year FE	Yes	Yes	Yes	Yes
SOE FE	No	Yes	Yes	Yes
Political Connection FE	No	No	Yes	Yes
Party cell FE	No	No	No	Yes

Notes: The unit of observation is a firm-year. The dependent variable is the number of government procurement contracts awarded to the firm that year, winsorized at the 0.1 and 99.9 percentiles. The independent variable is the firm's public alignment in the previous year, measured as the first principal component of Party-related phrase indicators. Both are standardized to have mean zero and standard deviation one within the estimation sample. Specification 1 includes year fixed effects. Specification 2 adds ownership fixed effects. Specification 3 adds political-connection fixed effects. Specification 4 adds Party-cell fixed effects. The estimation sample is held fixed at the most saturated specification. Standard errors are clustered by firm. Significance levels are indicated by * p<0.10, ** p<0.05, *** p<0.01.

Table A.11: Firm Performance Response to Unrest by Public Alignment

	(1) Revenue	(2) Profitability	(3) Labor Productivity	(4) Subsidies
$\text{Alignment}_{i,t} \times \text{Unrest}_{c(i),t+2}$	0.006	-0.018	-0.008	0.019
	(0.006)	(0.013)	(0.006)	(0.012)
$\text{Alignment}_{i,t} \times \text{Unrest}_{c(i),t+1}$	-0.008	0.021	0.016*	-0.019
	(0.007)	(0.013)	(0.009)	(0.016)
$\text{Alignment}_{i,t} \times \text{Unrest}_{c(i),t}$	0.037*	-0.006	-0.009	0.017
	(0.021)	(0.008)	(0.008)	(0.015)
$\text{Alignment}_{i,t} \times \text{Unrest}_{c(i),t-1}$	-0.021	-0.008	0.002	-0.016
	(0.013)	(0.010)	(0.007)	(0.019)
$\text{Alignment}_{i,t} \times \text{Unrest}_{c(i),t-2}$	0.017	-0.009	-0.003	-0.001
	(0.013)	(0.008)	(0.004)	(0.005)
Observations	34,639	34,419	34,370	28,710
R-squared	0.887	0.287	0.645	0.360
Firm FE	Yes	Yes	Yes	Yes
Year × Province FE	Yes	Yes	Yes	Yes
Unrest × SOE FE	Yes	Yes	Yes	Yes
Unrest × Political Connection FE	Yes	Yes	Yes	Yes
Unrest × Party cell FE	Yes	Yes	Yes	Yes

Notes: The unit of observation is a firm-year. Each column reports a different dependent variable: total revenue, winsorized at the 0.1 and 99.9 percentiles (column 1); profitability, defined as winsorized profit divided by winsorized revenue (column 2); labor productivity, defined as winsorized revenue per employee (column 3); and total government subsidies received by the firm (column 4). The independent variables of interest are interactions between the firm's public alignment, measured as the first principal component of Party-related phrase indicators, and labor unrest in the firm's city from $t-2$ through $t+2$. Because the outcome is measured in year t, the $t+1$ and $t+2$ interactions are the one- and two-year pre-period coefficients, whereas the $t-1$ and $t-2$ interactions are the one- and two-year post-unrest coefficients. Each column reports the most saturated specification: it controls for uninteracted unrest at all leads and lags, alignment, firm fixed effects, province-year fixed effects, and interactions of unrest at all leads and lags with SOE, political-connection, and Party-cell fixed effects. To facilitate interpretation, the dependent variable, the public-alignment score, and the unrest variables are standardized to have mean zero and standard deviation one within each column's estimation sample. Standard errors are clustered at the city level. Significance levels are indicated by * p<0.10, ** p<0.05, *** p<0.01.

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